Social Media Data Transparency for Information Integrity

Data Not Found

April 2026

About This Report

This report is the result of a collaboration between NetLab UFRJ, a research laboratory within the Federal University of Rio de Janeiro (Brazil), and the Minderoo Centre for Technology and Democracy at the University of Cambridge (United Kingdom).

Founded in 2013, NetLab UFRJ is one of Brazil's leading internet research laboratories. We are dedicated to investigating and analysing digital disinformation and its impacts in the country, producing empirical evidence to inform public debate and contribute to meaningful social impact.
Learn more: http://netlab.eco.ufrj.br

The Minderoo Centre for Technology and Democracy (MCTD) is an independent team of researchers at the University of Cambridge who are rethinking the power relationships between digital technologies, society, and the planet. We engage a broad set of stakeholders, helping to bring expertise and diverse voices to decisions about digital technologies and the future of artificial intelligence.
Learn more: http://www.mctd.ac.uk

Laboratório de Estudos de Internet e Redes Sociais

UFRJ UNIVERSIDADE FEDERAL DO RIO DE JANEIRO

MINDEROO CENTRE FOR TECHNOLOGY & DEMOCRACY

UNIVERSITY OF CAMBRIDGE

How to cite

Santini, R. M., Leal, H., Salles, D., Belisário, A., Mattos, B., & Pinho, D. (2026). *Data Not Found: Social Media Data Transparency for Information Integrity*. NetLab UFRJ & Minderoo Centre for Technology and Democracy. https://doi.org/10.17863/CAM.128975

Team

Research Directors

R. Marie Santini
NetLab UFRJ

Hugo Leal
Minderoo Centre for Technology & Democracy

Débora Salles
NetLab UFRJ

Project Coordinators

Adriano Belisário
NetLab UFRJ

Bruno Mattos
NetLab UFRJ

Danielle Pinho
NetLab UFRJ

Platform Assessments

Anita Cuteanu
Minderoo Centre for Technology & Democracy

Raphael Hernandes
Minderoo Centre for Technology & Democracy

Fernando Ferreira
NetLab UFRJ

Felipe Grael
NetLab UFRJ

João Gabriel Haddad
NetLab UFRJ

Lucas Murakami
NetLab UFRJ

Rafael Cardoso
NetLab UFRJ

Vinicius Branco
NetLab UFRJ

Editorial Review and Design

Felipe Loureiro
NetLab UFRJ

Rowena Jameson
NetLab UFRJ

Gina Neff
Minderoo Centre for Technology & Democracy

Ann Kristin Glenster
Minderoo Centre for Technology & Democracy

Acronyms 07

Abstract 08

Foreword 09

Executive Summary 12

Key Findings 14

Key Recommendations 16

Introduction 19

The Data Transparency Crisis 20

Report Overview 23

Transparency and Information Integrity 25

Regulating Data Transparency in Social Media 31

European Union 33

Brazil 35

United Kingdom 37

Materials & Methods 39

Scope of Analysis 41

Scoring Criteria 43

Results 45

User-Generated Content Data Transparency Assessments 48

Advertising Content Data Transparency Assessments 52

Recommendations 57

Social Media Companies 59

International Governance Bodies 62

Regional Regulation 63

Public-Interest Researchers and Institutions 66

Bibliography 68

Terminology 77

Acronyms

AI
Artificial Intelligence

API
Application Programming Interface

DSA
Digital Services Act

EU
European Union

GUI
Graphical User Interface

MCL
Meta Content Library

MCTD
Minderoo Centre for Technology and Democracy

OECD
Organisation for Economic Co-operation and Development

OSA
Online Safety Act

TTPA
Transparency and Targeting of Political Advertising (EU Regulation)

UGC
User-Generated Content

UK
United Kingdom

UN
United Nations

USA
United States of America

UNESCO
United Nations Educational, Scientific and Cultural Organization

UFRJ
Universidade Federal do Rio de Janeiro (Federal University of Rio de Janeiro)

VLOP
Very Large Online Platform

VLOSE
Very Large Online Search Engine

Abstract

Social media platforms play a central role in contemporary public discourse, yet remain largely inaccessible to independent scrutiny. Transparency of social media data is essential for ensuring information integrity. Limited access to such data undermines evidence-based policymaking, democratic accountability, efforts to mitigate online harms, and the effective enforcement of laws across jurisdictions.

This report presents the ***Social Media Data Transparency Index***, a systematic, first-of-its-kind assessment of data access conditions across 15 major social media platforms across the different regulatory contexts of the European Union, the United Kingdom, and Brazil. Developed collaboratively by **NetLab UFRJ**, at the Federal University of Rio de Janeiro (Brazil), and the **Minderoo Centre for Technology and Democracy** at the University of Cambridge (United Kingdom), the Index evaluates platform transparency in terms of the ability to access and analyse public user-generated content (UGC) and advertising data using a framework grounded in data quality principles.

We find transparency is still the exception rather than the norm. This is regardless of the regulatory frameworks in place. Many widely-used platforms provide little or no access to data in any of the regions assessed, while others impose restrictions that prevent independent scrutiny, including public-interest research, journalism, and oversight by civil society and government regulators. The lack of transparency in user-generated content data makes it difficult to track information flows, hindering efforts to monitor disinformation and safeguard information integrity. In advertising, opacity in platform data is especially harmful, exposing users to risks by undermining consumer protection and limiting market actors' ability to make informed, auditable decisions. Together, these gaps weaken the capacity of researchers, regulators, and businesses to act effectively, with tangible societal and financial risks.

The results suggest that regulatory frameworks, particularly the European Union's Digital Services Act, can improve data access conditions, especially for advertising data, but they are not sufficient for ensuring effective implementation or compliance. Brazil lacks a dedicated regulatory framework for platform transparency and consistently shows the lowest levels of data access. The United Kingdom, on the other hand, benefits indirectly from European Union legislation, reflecting the so-called "Brussels Effect".

The report provides recommendations for coordinated international action by social media companies, governance bodies, regional policymakers, and public-interest researchers to improve transparency of public user-generated content and advertising data. These recommendations focus on supporting public-interest research and addressing gaps that drive regional and platform-level asymmetries. They also call for stronger enforcement of transparency standards, the development of interoperable data access protocols, and efforts to reduce disparities that undermine global information integrity.

Foreword

There is a paradox at the heart of the social media age. Platforms that billions of people rely upon to share information, make markets, organise civic life, and participate in democratic debate are themselves almost entirely shielded from independent public scrutiny. They know a great deal about us. We know almost nothing about them. This report, and the *Social Media Data Transparency Index* it presents, is a rigorous attempt to map that asymmetry—and to make the case that closing it is a matter of urgency.

Transparency is the precondition for accountability. Without the ability to observe how content flows across platforms, how advertisers target audiences, and how information operations exploit the interplay between organic and paid content, we cannot hold enormously powerful social media companies to account. Regulators cannot enforce the law. Researchers cannot produce the evidence base that democratic policymaking requires. Journalists cannot investigate. And citizens cannot make informed choices about the digital environments that shape so much of their lives.

The team at the Minderoo Centre for Technology and Democracy (MCTD) was founded on precisely this conviction: that rethinking the power relationships between digital technologies, society, and the planet requires evidence. In the context of social media, evidence means access to data that companies increasingly hide inside online "walled gardens". That is why MCTD has consistently placed the access to social media data in the public interest at the centre of its research agenda—not as an end in itself, but as the essential infrastructure for democratic oversight and public accountability.

This report is the result of a collaboration that I am genuinely proud of. The research team at NetLab UFRJ, led by Professor R. Marie Santini, has spent years building one of the most rigorous and consequential programmes of empirical research on platform transparency, disinformation, and digital governance anywhere in the world. Their particular expertise—in the architecture of data access, the mechanics of advertising opacity, and the real-world consequences of platform non-compliance—is unmatched. The partnership between NetLab and MCTD combines complementary strengths: NetLab's deep technical and empirical grounding in the Brazilian and Latin American context, and MCTD's public-facing translational research on the governance challenges that confront democratic societies across Europe and beyond.

What makes this report particularly important is its ambition to paint an international comparative picture. The platforms under scrutiny here are not local actors operating within tidy jurisdictional boundaries, but global infrastructure. When Meta, TikTok, or X make data access policies, their decisions ripple across every country that relies on their services. A transparency failure in Brazil—where this report finds consistently the lowest levels of data access—is not just a Brazilian problem. The asymmetry this report documents, in which the same companies offer meaningfully greater access in markets where regulation compels them to do so, is a choice that social media platforms have made and our governments have acquiesced to. This report names it as such.

Researchers working in this field know the contours of the landscape we report

here, but not the specific topography. The data transparency crisis that is the subject of this report has been building for over a decade. In the early years of social media research, platforms were relatively open: APIs were accessible, data was available, and a generation of computational social scientists built an evidence base on phenomena from disinformation to radicalisation to public health communication that simply could not have existed otherwise. That openness, always imperfect, has been systematically eroded. The Cambridge Analytica scandal accelerated a closing of the gates, for it was not-for-profit public interest research that paid the price, not the commercial actors whose misuse had prompted the crackdown. The erosion has continued: CrowdTangle discontinued, APIs paywalled, access requests denied or unanswered, transparency tools deployed that return no results. Our own researchers' experience in the course of producing this report—including having an access request to X's API under the Digital Services Act definitively rejected after months of follow-up—is itself testimony to the scale of the problem.

Data transparency is the means by which societies can hold powerful institutions to account. The online advertising market, now estimated at USD 650 billion globally, runs largely on unauditable, self-reported metrics. Electoral campaigns are conducted on platforms where neither regulators nor researchers can independently verify the reach, content, or targeting of political advertising. Disinformation networks exploit the interplay between organic and paid content in ways that are structurally invisible to outside observers. Scam advertisers operate without scrutiny. These risks are the documented reality of the information environment.

The twelve recommendations set out in this report are addressed to social media companies, international governance bodies, regional policymakers, and the research community itself. They are grounded in evidence and calibrated to what is technically and institutionally feasible. They make practical and operationalisationable demands for what the United Nations itself has called for: meaningful transparency, enforced by binding standards, that enables independent research and democratic oversight. This is what societies need to guarantee information integrity.

I am grateful to every member of the research team who contributed to this work, both at NetLab UFRJ and at the Minderoo Centre for Technology and Democracy. The effort involved in rigorously assessing 15 platforms across three jurisdictions, negotiating access requests, testing interfaces, and validating findings over months of collaborative work is immense, and the quality of the scholarship here reflects it. I am particularly grateful to Professor Santini and Dr Hugo Leal for their intellectual leadership for this collaboration from its outset.

The online world does not have to live in a walled garden. But making a different future will require the sustained pressure of rigorous research, coordinated advocacy, and, ultimately, the political will to insist that accountability is not optional. That work is what this report—and the partnership that produced it—is for.

Professor Gina Neff CAMBRIDGE, APRIL 2026

Minderoo Centre for Technology and Democracy University of Cambridge

Professor of Responsible AI, Queen Mary University of London

Executive Summary

The *Social Media Data Transparency Index* provides a detailed assessment of the deepening transparency crisis that undermines public-interest research, democratic oversight, and the credibility of the advertising market in an increasingly online world. Research teams at **NetLab UFRJ**, at the Federal University of Rio de Janeiro (Brazil), and the **Minderoo Centre for Technology & Democracy**, at the University of Cambridge (United Kingdom), collaborated to develop this policy report in order to contribute to the evidence on social media data transparency. This research builds on work initially developed by NetLab UFRJ in 2024 to evaluate access to social media data in Brazil. This systematic assessment was expanded to 15 major platforms across three regions—the European Union (EU), the United Kingdom (UK), and Brazil—which were ranked on a 100-point scale, from **Meaningful** to **Negligible**, based on access to user-generated content and advertising data.

Across the jurisdictions and platforms analysed, data transparency remains uneven and frequently limited. These patterns fall short of the United Nations (UN) recommendations and raise serious concerns for information integrity and democratic governance. The UN identifies information integrity as a foundational pillar of a healthy digital ecosystem and recognises transparency as one of its core principles. We recommend that decision-makers ensure meaningful transparency by enabling access for trusted independent research organisations, as well as journalists, civil society, and international organisations[1]. Our methodology defines meaningful transparency as openly accessible, well-documented data access resources with minimal barriers to data extraction. This report shows how current practices fall short of these standards and enables comparison across regions with differing regulatory frameworks.

Dedicated data transparency regulation is improving data access conditions. Among the regions assessed, the EU stands out for having a dedicated transparency framework for social media platform data. The Digital Services Act (DSA) has introduced a landmark regulatory framework for user-generated content (UGC) and advertising data access. Our evaluation shows improved data access conditions in the EU, particularly regarding advertising transparency. Brazil, like much of the Global Majority, lacks an equivalent data access mandate and consistently scores lowest, as platform policies remain largely voluntary. On average, the United Kingdom scores comparatively closer to the EU. While the UK's Online Safety Act establishes oversight of platforms, it relies largely on case-by-case assessments by regulatory authorities and does not provide a comprehensive transparency framework for social media data access. We hypothesise that the country may be indirectly benefiting from the so-called "Brussels Effect", whereby EU regulations shape platform practices beyond the Union's borders, even where no such regulations are in force.

However, regulatory differences matter less than expected: across platforms, regions with established frameworks often face obstacles similar to those with no regulation at all. Regulation

[1] Throughout this report, public-interest research is understood broadly, in line with the UN framing, and is not limited to academic, university-based activities.

alone does not guarantee transparency or effective data access mechanisms, as implementation continues to depend heavily on platform discretion in interpreting and operationalising these obligations. Significant challenges to implementation and enforcement persist, even in the EU, which has led the way in adopting comprehensive data access regulations. The absence of common access and data quality protocols, combined with weak enforcement, limits the impact of the DSA. Our assessment aligns with prior research showing that platforms such as X, Snapchat, and Pinterest—despite falling within the scope of the DSA—still fail to meet minimum transparency standards. This underscores a broader pattern: formal regulatory frameworks do not necessarily translate into functional and reliable data access for public-interest research.

Data transparency is not an end in itself, but part of a broader transparency framework that enables accountability and democratic oversight of social media platforms. Companies, policymakers, and researchers must work to ensure that social media data access is transparent, legally compliant, and usable for public-interest oversight. Persistent gaps in how transparency and data access are operationalised must be addressed to safeguard the integrity of online information ecosystems. Doing so will require coordinated action from different actors. This report is a step in that direction.

Key Findings

01

Social media data transparency remains poor across the EU, Brazil and the UK. Our findings show that widely used services provide no data transparency mechanisms for either user-generated content (UGC) or advertising, including both application programming interfaces (APIs) and graphical user interfaces (GUIs), across any of the assessed regions. Reddit, for instance, provides programmatic access to public UGC data but not to advertising data in any region assessed. X (formerly Twitter) offers access to public UGC data, but subject to prohibitive pricing structures, while its advertising repository, available only in the EU, returns no results. While Meta provides **Meaningful** access to advertising data in the UK, its UGC transparency score was deemed **Negligible** across all analysed regions. Platforms such as TikTok and LinkedIn offer, at best, **Deficient** or **Limited** transparency for both UGC and advertising data.

02

Disparities in data access around the world reflect a compliance-driven model of selective transparency. Transparency conditions vary across jurisdictions, particularly for advertising data, with stronger access observed in the UK and the EU, regions with more robust regulatory frameworks—or, in the case of the UK, potentially reflecting regulatory spillovers. Brazil, which lacks a dedicated framework for platform transparency, consistently records the lowest levels of data access. On average, the UK scores comparatively closer to the EU.

03

Robust mechanisms for accessing public user-generated content data are scarce. Bluesky and YouTube were the only platforms whose public APIs were deemed **Meaningful** in our analysis, consistently across all regions assessed. They do not differentiate the public content that can be retrieved, and their functionalities performed well in our tests. X, Telegram, and Reddit present limitations in their APIs—most notably, in the case of X, the costs associated with access. Public UGC data from Facebook and Instagram can only be accessed through the Meta Content Library, whose API is restricted to secure and controlled environments and does not allow for the extraction of disaggregated data to users' own infrastructure. While platforms such as LinkedIn and TikTok offer researcher APIs under the DSA, Pinterest and Snapchat only accept data requests from researchers, which is considered insufficient under our methodological approach.

04

Existing tools for accessing and exporting advertising data remain too limited for meaningful scrutiny. The reliance on APIs and GUIs with limited search capabilities, combined with restricted access to granular data, prevents researchers from effectively assessing targeting practices, advertising reach, and the broader societal impacts of online advertising. Platforms such as Meta, LinkedIn, and TikTok allow ads to be queried using keywords, but others, including X, Google (YouTube), and Pinterest do not, limiting data discovery and retrieval to searches based on advertiser names. In addition, platforms disclose reach and targeting data in broad value ranges, further restricting both independent scrutiny and advertisers' ability to accurately measure campaign performance and return on investment.

Key Recommendations

Social Media Companies

R1

Ensure meaningful universal access to all advertising data. Platforms should provide free, programmatic, and open access to advertising repositories containing the full dataset of ad data, made available through both APIs and GUIs, that meet robust data quality standards and do not differentiate access based on ad content. This is essential for equipping researchers and regulators to identify risks, evaluate compliance, and enforce accountability in online information ecosystems.

R2

Ensure meaningful universal access to public user-generated content data for public-interest uses. Platforms should provide free, programmatic, and non-discretionary access to public UGC data outside controlled or secure environments, ensuring it meets clear standards of completeness and quality and is made available through both APIs and GUIs to accommodate different levels of technical capacity and expertise. This is central for ensuring that social media platforms meet their legal requirements and that citizens and decision-makers can access critical information that they need.

R3

End selective and fragmented data transparency practices. Platforms should align their global UGC and advertising data transparency practices with the most open standards, rather than relying on minimal or jurisdiction-specific compliance, thereby minimising selective transparency driven by uneven regulatory oversight.

R4

Provide vetted researchers with access to non-public data within the limits defined by democratic regulation and oversight. Access to non-public platform data is essential for monitoring platform-enabled risks and harms, as well as for independently auditing platforms' self-reported transparency metrics. Under Article 40 of the DSA, such access is formally regulated. Although this report does not examine access to non-public data, we emphasise the need for robust researcher vetting processes and secure protocols to enable its responsible availability.

R5

Provide access to data on moderated and removed content. This data, for both UGC and advertising, is crucial for understanding threats to information integrity and assessing how platforms address and moderate them. Companies should retain such information in auditable databases, rather than erasing or losing it.

International Governance Bodies

R6

Advance the information integrity agenda through data access and transparency standards. Data transparency mechanisms should move from self-regulated commitments to legally binding regulatory frameworks that enable effective oversight. These should be supported by the establishment and promotion of international principles and standards governing the technical, ethical, and operational dimensions of data access.

Regional Regulation

R7

European Union: Fully enforce Articles 39 and 40 of the Digital Services Act. The European Commission should ensure the effective implementation of advertising and public UGC data access mechanisms through standardised protocols and be prepared to impose meaningful penalties—including substantial fines and, where necessary, temporary suspension of services—for non-compliance. Our report suggests doing so may have an impact beyond the EU.

R8

Brazil: Consolidate existing legal principles into a dedicated transparency framework for platform data access. Brazilian authorities should develop a dedicated framework for social media data transparency, building on lessons from past efforts, consolidating principles from existing data protection, consumer protection, and child protection laws, and leveraging the country's multistakeholder internet governance approach.

R9

United Kingdom: Move from case-by-case oversight toward a dedicated transparency framework for platform data access using the Online Safety Act framework. The UK should adopt a proactive framework for platform data transparency, drawing on lessons from the EU's DSA and helping to bridge the gap between UK and EU-based researchers and institutions.

R10

Brazil and the United Kingdom: Clarify the legality and ethical use of web scraping for public-interest research. Brazilian and UK authorities should follow the DSA's provisions on publicly available data and explicitly support the legal and ethical use of web scraping for public-interest research and oversight, particularly where platform-provided transparency is insufficient.

Public-Interest Researchers and Institutions

R11

Promote international research consortia and foster interdisciplinary collaboration. Researchers and research institutions could pool knowledge and technical resources through multinational, cross-sector, and interdisciplinary collaboration, actively supporting regulatory efforts by contributing expertise, documenting successes and failures, and working across jurisdictions to strengthen the evidence base for global platform governance. We call on research funders to support cross-national research initiatives.

R12

Develop monitoring strategies for low-access settings. Researchers should develop open-source tools and design technical and methodological approaches that can be readily deployed in contexts with severely restricted data transparency and limited resources, such as browser-based tools, cloud-based data pooling, and crowdsourced data donations.

Introduction

Chapter 01

The Data Transparency Crisis

Social media platforms are responsible for a transparency crisis that is threatening the integrity of our information ecosystem. Although participation in social media platforms is open to billions of users, companies have effectively closed many of the avenues for effective scrutiny. Online services appropriate public data for private gain while keeping public-interest research in the dark. A regime designed for information asymmetry and operational opacity leaves serious questions about social life unanswered: who is orchestrating the online disinformation campaigns that are fostering an unprecedented wave of democratic backslide? How did infodemics become one of the major threats to public health? What is the relationship between individual acts of online abuse against young women and emergent misogynistic cultures?

Without access to data, researchers cannot address major areas of social change, policymakers lack evidence for decision-making, and society fails to have the information it needs to act. Social media companies are resisting independent public-interest scrutiny by researchers, journalists, policymakers, and citizens. Our increasingly datafied societies operate in walled gardens of an online media ecosystem where data are not found.

The landscape was not always this bleak. During an initial period of market expansion, social media companies opened their platforms to programmatic interaction and interoperability. They introduced Application Programming Interfaces (APIs)[2] as a mechanism to enable the communication between different applications. The relationship was mutually beneficial. The API owners offered new services on their platforms without significant costs and the independent developers gained visibility and reach. Data access, albeit limited, was part of this tradeoff.

The so-called big data revolution benefited from access to social media data. Over more than a decade, we witnessed an unprecedented growth in the quantity and quality of scientific and public-interest research with big data. The promise of "Computing Humanity" (Ledford, 2020) did not come without its significant risks and challenges. The Cambridge Analytica scandal, a case of personal data misuse for commercial and political purposes enabled by Facebook's API, showed that programmatic interaction was a delicate ecosystem (OECD, 2019) in need of robust regulation and democratic oversight (European Parliament, 2018). Instead, social media companies used their power to redefine the terms of engagement. Public-interest research ultimately bore the cost as

[2] Further clarification on the terminology used is provided in the *Terminology* section at the end of this report.

public APIs were deprecated and personal data increasingly locked within digital enclosures.

Twitter (now X), long regarded as one of the most research-friendly social media platforms, aggressively monetised its API in 2023. The costs are prohibitive for most research institutions (Coalition For Independent Technology Research, 2023). For example, a landmark 2018 study on the spread of true and false news drew on a dataset of over 4.5 million posts, along with an additional 10 million tweets used for modelling purposes (Vosoughi et al., 2018). Until February 2023, academics could collect such volumes of data for free. Under the pricing structure introduced thereafter, researchers would need the enterprise plan at USD 42,500 per month (X, n.d.-a), while the pay-per-use tier introduced in 2026 would be even more onerous, with estimated costs of USD 72,500, and technically unfeasible due to a cap of 2 million post reads per month (X, n.d.-b).

In 2024, Meta shut down CrowdTangle, a formerly independent media insights tool the company had acquired and made available to academics and journalists, replacing it with the Meta Content Library (MCL). Our report shows that this is a poorly performing tool only available to researchers pre-approved by Meta. After an arbitrary approval process, researchers must navigate closed virtual machine systems that prevent data exports. Other platforms promoted a narrow set of official data-access solutions, such as the TikTok Research API, which is restricted to researchers based in the United States and Europe (Pearson et al., 2025).

The rise of data scraping for artificial intelligence (AI) training further limits independent research. Large language models and other generative AI systems increasingly rely on vast amounts of data derived from texts, images, and videos published on online platforms. Social media companies—including Reddit and X—resist third-party uses of what they frame as "proprietary data" to train new models without compensation (Joseph, 2025; Murphy et al., 2025). As a consequence, access to social media data by researchers, with or without permission, has become increasingly difficult and contentious, as the same data access mechanisms developed for scholarly inquiry can also be leveraged by large AI companies to train commercial tools.

Business models based on the monetisation of personal data thrived while public-interest research with public data from social media platforms suffered. Without adequate data transparency, our ability to know what happens to people online is subject to the choices made by social media companies.

Debates over data access have increasingly extended beyond user-generated content (UGC) to encompass publicly available advertising data. Transparency in advertising practices is essential for scrutinising paid content, a cornerstone of the multi-billion dollar digital advertising market—estimated at USD 650 billion in 2025 (Precedence Research, 2025). Access to advertising data remains constrained and has received less attention compared to user-generated content. This asymmetry is not incidental: disclosing advertising data opens up platforms' commercial interests and their business model to scrutiny. There are strong incentives to limit public availability to this data and curtail external oversight, and these incentives are misaligned with the public's need to know about advertising online.

For example, revelations about the role of political advertising in shaping public opinion and influencing democratic processes have cast a light on the importance of advertising transparency in regulatory and academic debates. These concerns stem from the same electoral and public opinion scandals that led platforms to restrict access to UGC data, especially those surrounding the 2016 US elections (Leerssen et al., 2019). At the same time, the sophisticated microtargeting capabilities offered to advertisers have been widely used in disinformation campaigns, influence operations, and in online scams that disproportionately target at-risk individuals (Horwitz, 2025; Santini et al., 2025b; Seers et al., 2026).

Major platforms responded to mounting public pressure and successive crises by increasing advertising transparency for the public, authorities, and researchers, partly as a strategy to pre-empt binding regulatory frameworks and mitigate any backlash. Google, Meta, and X/Twitter were among the first to develop self-regulated ad repositories, or ad libraries, web-based tools often accompanied by APIs for data exploration and extraction, designed to enable greater scrutiny of ads served on major platforms. These systems typically provide information on targeting criteria, audiences reached, ad spending, delivery periods, and, in some cases, content moderation actions (Leerssen et al., 2019; Mozilla Foundation, 2019).

Researchers and civil society organisations widely consider ad repositories inadequate and insufficient. These archives impose significant constraints on bulk access, historical depth, and analytical functionality: datasets are often incomplete or inconsistent, interfaces are difficult to navigate, and information is provided in formats that hinder large-scale analysis and systematic reporting (European Commission, 2018; Rieke & Bogen, 2018; Santini et al., 2024; Who Targets Me, 2025).

Even if political advertising has been treated as the primary focus of these tools, given its disproportionate impact on public opinion and social institutions, self-regulated ad repositories have left other forms of advertising largely, and sometimes entirely, opaque. Advertising can play a significant role in funding extremist content and disinformation. For example, the digital advertising watchdog Check My Ads found obscure advertising played a role in the 2024 racially-motivated riots in Southport, UK (Check My Ads Institute, 2025) Moreover, platforms diverge significantly in how they define and operationalise "political advertising": some adopt narrow, election-focused criteria, while others employ broader frameworks that encompass social issues (Sosnovik & Goga, 2021).

Political ad categorisation relies on self-classification by advertisers, who may mislabel their content, intentionally or not, thereby directly affecting what is made transparent. More fundamentally, assigning the responsibility of determining what constitutes political content worldwide to a small number of private companies poses significant risks, effectively turning these platforms into arbiters of electoral campaigning and political discourse across jurisdictions. Given the difficulty of drawing clear boundaries between political and non-political advertising, many explicitly political ads remain uncategorised, while non-political ads are sometimes incorrectly labelled as political, weakening the effectiveness of platform

transparency mechanisms (European Partnership for Democracy, 2020; Leerssen et al., 2019; Le Pochat et al., 2022; Santini et al., 2025b).

From pharmaceutical and health disinformation to online financial scams, there are areas where the lack of information about online advertising harms people. Easily searchable databases of ads and advertisers could be a key step in protecting users, preserving the public's trust in online advertising systems and improving the fairness and transparency in online ad markets (Schiffrin et al., 2026). The UK's Online Safety Act provides some duties around advertising, but "falls short of delivering a comprehensive regulatory framework, leaving some gaps in coverage, enforcement, and oversight of third-party ad delivery systems" (Woods & Antoniu, 2025).

Regulatory experiences have already shown that tools to monitor online advertising content can—and should—be substantially improved. Understanding the commercial operations that underpin the economics of large social media platforms is a matter of public interest which cannot remain obscured by transparency tools that fall short of their stated goals.

Report Overview

In the following sections, we first examine the intrinsic relationship between platform transparency, oversight, and the broader debate on information integrity, as primarily outlined by the United Nations. While information integrity is currently threatened by major disruptions in communication and society, transparency can play a fundamental role in addressing the asymmetries that enable these dynamics.

Next, the section *Regulating Data Transparency in Social Media* examines the regulatory frameworks aimed at enhancing platform transparency and accountability for the societal damage they enable across the three regions covered by our analytical framework: the European Union (EU), Brazil, and the United Kingdom (UK). While the Digital Services Act (DSA) in the EU establishes provisions governing access to public UGC and advertising data—primarily to address platform-enabled systemic risks—Brazil and the UK do not currently have dedicated legal frameworks for data transparency.

The section *Materials & Methods* presents our methodological approach, focusing on the evaluation framework used to assess transparency in public UGC and advertising data across the three aforementioned regions, as applied to 15

major social media platforms. We adopt a data quality–focused approach, assessing not only whether platforms provide publicly available data through dedicated transparency mechanisms, but also the conditions under which such access is granted.

Finally, we present the results of applying this framework, showing that meaningful transparency is the exception rather than the norm across all three analysed jurisdictions, for both public UGC and advertising data. Although the DSA contributes to reducing platform opacity in the EU, it still falls short of what we define as meaningful transparency, due to limitations in enforcement and the lack of clearly defined common standards. These findings are followed by a call for multistakeholder action, set out through a series of recommendations addressed to key actors, including platforms, international governance bodies, regional regulators and policymakers, and researchers.

The online edition[3] of this report also provides a full description of our methodological approach, with references outlining how transparency is conceptualised within the information integrity framework.

[3] Available at https://netlab.eco.ufrj.br/transparency

Transparency and Information Integrity

Chapter 02

Online platforms shape public debate, access to information, and modes of expression in ways that are often opaque and misaligned with the public interest. Given their substantial societal influence, effective scrutiny and accountability are urgent—and both rely on access to platform data. Researchers and civil society have used such data to uncover how harmful phenomena operate within these ecosystems at an unprecedented scale, and to assess how platform features may generate or amplify societal harms, particularly for vulnerable groups. These harms include coordinated disinformation campaigns and foreign influence operations that spread conspiracy theories and manipulate elections, hate speech targeting minorities and other social groups, and the recruitment and mobilisation of terrorist or neo-Nazi actors, often reinforced by platform monetisation mechanisms (Benkler et al., 2018; Bradshaw & Howard, 2019; Davey & Ebner, 2019; Marwick & Lewis, 2017; Tucker et al., 2018).

Exposing these dynamics through critical research has increased the risks to platforms' reputations and invited regulatory oversight, making those responsible for the calls for transparency clear targets of platform discontent (Bruns, 2019; Tromble, 2021). Several platforms have discontinued previously available data access tools, such as APIs, often replacing them with weaker and less effective alternatives (Bruns, 2019; Freelon, 2018). Recent evidence reinforces this concern: over 60% of researchers surveyed by the Coalition for Independent Technology Research reported facing barriers to accessing the data essential for studying how technology affects society (Coalition for Independent Technology Research, 2025).

Data Not Found, Democracy at Risk: Online manipulation and electoral upsets in Romania

The 2024 presidential election in Romania brought to light the dangers of social media data opacity. Călin Georgescu, a far-right, pro-Russian candidate dealing in conspiracy theories and nationalist tropes, went from political invisibility to electoral success through a seemingly well-orchestrated online blitz. Two weeks before the first round, Georgescu was languishing in the opinion polls, struggling for political relevance. On the night of 24 November, he was celebrating a shocking victory and preparing to face the liberal Elena Lasconi in the runoff (Henley, 2024).

The electoral upset was followed by accusations of campaign fraud and allegations of a Foreign Information, Manipulation and Interference operation led by Russia. In the wake of the public outcry and supported by intelligence services reports uncovering financial irregularities and a wave of foreign cyberattacks (Rainsford, 2024), the Romanian Constitutional Court annulled the first round of the presidential election just two days ahead of the second round. This decision plunged Romania in a political turmoil for months, threatening the integrity of the democratic process in the country.

What happened? Regardless of the discussions surrounding the legitimacy and legality of the decisions to cancel the election and ban Georgescu, how could an extremely online fringe political figure become a mainstream national candidate vying for the presidential election in a matter of weeks without raising any alarm bells? Under the current social media data access restrictions, several pieces of information have been assembled but the puzzle has not been solved. Since November 2024, Romanian and international organisations have been combing through the dearth of data in an effort to inspect the social media data and algorithmic black boxes. These exercises of digital forensics found traces of a sophisticated online architecture supported by foreign actors (Nuţu et al., 2025; Olari, 2024).

Georgescu sidelined the traditional electoral campaign and relied almost exclusively on the network effects of social media platforms, like Telegram, Facebook and TikTok. According to some investigations (Nuţu et al., 2025; Olari, 2024), the candidate's social media engagement grew exponentially in the weeks preceding the election, which allowed him to reach more young voters in the country and in the Romanian diaspora. Although Georgescu's campaign claimed zero spending, the Romanian intelligence service claimed to have found digital footprints of foreign manipulation and irregular campaign funding used to promote the candidate on TikTok (Rainsford, 2024).

In yet another demonstration of the shortcomings of online self-regulation, TikTok had decided to ban political advertising in 2019 (Chandlee, 2019). This decision, however, did not prevent the platform from becoming the main hub for an online campaign that undermined the democratic process in a EU member country. Under the DSA, the European Commission issued a retention order to TikTok, instructing the platform to preserve all data related to systemic risks against democratic electoral processes (European Commission, 2024b). Too little, too late. Only the implementation and enforcement of robust frameworks that ensure access to user-generated and advertising data will allow the effective monitoring and facilitate the prevention of online manipulation and fraud.

Researcher access to data is particularly important for understanding both the problems and potential solutions associated with global technology harms. The global scientific community has identified issues such as child exploitation, technology-facilitated harms, and climate misinformation as areas where constraints on access to platform data are limiting our ability to generate evidence and respond effectively to these challenges (Pothong et al., 2026).

Against this backdrop, the United Nations (UN) and other international bodies have situated data transparency within the emerging debate on so-called information integrity. Broadly speaking, information integrity refers to information that is reliable, accurate, and accessible to all, that empowers people to engage in public life and debate, and is shaped by governments, technology companies, media organisations, civil society, and individuals alike (United Nations, 2025). More broadly, any factor that undermines users' ability to access accurate, consistent, and reliable information, free from manipulation and falsehoods, constitutes a threat to information integrity (European Parliament, 2024; United Nations, 2024).

Achieving information integrity requires mitigating risks such as disinformation, hate speech, and toxic discourse in the wider sense. The UN has identified some of these risks as major global vulnerabilities that can arise from the malicious use of technology. Mitigation measures, in turn, involve identifying adversaries—including individuals, groups, organisations, and both State and non-State actors—along with their tactics and objectives, while carefully considering potential outcomes. These disruptions threaten democratic governance, human rights, peace, and effective responses to environmental and health crises, hindering public understanding, informed debate, and evidence-based decision-making (United Nations, 2025).

Building an information ecosystem with integrity requires safeguarding freedom of expression and ensuring the free, safe, and inclusive flow of information, while respecting individual privacy. In this context, the UN's principles for strengthening information integrity, set out in a call for multistakeholder action, focus on key pillars such as supporting independent, free, and pluralistic media, empowering citizens, and, most relevant to this study, ensuring that platforms provide transparency guarantees for public-interest research (United Nations, 2024). Notably, Research is one of the pillars of the UN's 3R approach to information integrity, alongside Response (Prevention, Protection, Mitigation, Recovery) and Risk (Assessment) (United Nations, 2025).

Access to data for public-interest research offers a way to bridge the gap between platforms and citizens by addressing a longstanding asymmetry: while platforms know a great deal about their users, users know very little about social media dynamics. Research based on online platform data ultimately enables measurements for platforms' impacts on society through observing human behaviour in real-world environments (Marres, 2017; Rogers, 2009; van Atteveldt & Peng, 2018).

The potential of such investigations to inform the design and revision of public policies should not be underestimated. The UN's Information Integrity Theory of Change broadly argues that research based on open data fosters public awareness and evidence-based insights, which in turn support more informed decision-making (United Nations, 2025). The online version of this report[4] provides a list of key references from international bodies that elucidate the role of transparency within the information integrity framework, with recommendations and guidelines that can act as a beacon for policy decision-making.

Moreover, the UN argues that debates on data access and transparency must be grounded in robust data quality principles to support accountability and informed policymaking, particularly in light of enduring power asymmetries and emerging risks, such as the rising use of generative AI for public-opinion manipulation (United Nations, 2024). Indeed, data quality is a central concern in data governance and lifecycle management. Making data available is only a baseline requirement: it must also meet a set of quality criteria that determine whether it is suitable for specific analytical or policy purposes

[4] Available at https://netlab.eco.ufrj.br/transparency

Data not Found, Investment at risk: Is digital advertising a racket?

Digital advertising is an example of a perfectly imperfect information market. Advertisers struggle to reliably estimate return on investment, while users are reduced to monetised data points. The market is controlled by online services and social media platforms selling virtual spaces and algorithms that promise microtargeted brand-user connection. Advertisers and investors continue to believe in the efficiency of this unverifiable mechanism.

Evidence has long cast doubt on these assumptions. In 2021, Forbes reported cases of companies withdrawing from digital advertising without measurable losses in sales or revenue (Fou, 2021). Earlier, a large-scale experiment by eBay Research Labs found that paid search advertising produced "no measurable short-term benefit" (Blake et al., 2015), prompting the company to scale back its investment.

Oblivious to the warning signs, the digital marketplace continued to expand. In 2017, the consumer-goods giant Procter & Gamble cut USD 200 million in digital ad spending after identifying significant limitations in targeting effectiveness. Marc Pritchard, the Chief Brand Officer of Procter & Gamble, made an illuminating declaration: "transparency [shined] a spotlight on reality and we learned valuable lessons which are driving profound change" (Cavale, 2018).

These market experiments by two leading players of the new and old economies unveiled deep-rooted structural problems: underneath the inflated claims of reach, impressions and microtargetting, lies an inefficient and often fraudulent market (Schiffrin et al., 2026) that has grown ever more opaque.

At the same time, a growing "attention crisis" further undermines the model. Users increasingly rely on ad-blocking technologies, while shifts toward AI-mediated search and interaction reduce the visibility and relevance of traditional ads. As more online activity takes place through AI systems rather than search engines or websites, the link between ad exposure and consumer behaviour becomes even more tenuous.

The near-future looks grim. At its most farcical, digital advertising risks devolving into a synthetic loop in which AI-generated ads are delivered primarily to automated bots. At its most serious, it could resemble less a marketplace and more a financial racket. This trajectory underscores the urgent need for greater transparency. Technical complexity and platform opacity should not preclude independent scrutiny or accountability.

Advertisers, investors, researchers, and the public must have meaningful access to data on how digital advertising systems operate, as well as robust safeguards against abusive practices. Moving toward a more symmetrical information environment is therefore essential. Data transparency in digital advertising is not only a matter of market efficiency, but a financial imperative and a fundamental consumer right.

(Batini & Scannapieco, 2016; Benson, 2019; Mahanti, 2019). In the context of social research, ensuring data quality is essential for guaranteeing the reliability and reproducibility of studies, as well as enabling generalisations (Srivastava & Mishra, 2021).

In fact, many researchers have previously questioned the quality of the data that platforms make available. For instance, researchers have long raised concerns about which data is made available, particularly regarding sampling and representativeness (Tufekci, 2014). Data made available through APIs and advertising repositories may also become unavailable or degrade over time, while researchers frequently face challenges related to outdated metrics and incomplete information (Angus et al., 2024; Edelson, 2020; Zubiaga, 2018). We structure our analysis around established data quality dimensions, allowing us to assess not only whether data is made available, but also the conditions under which access is granted, as detailed in our methodological complement, available online[5].

[5] Available at https://netlab.eco.ufrj.br/transparency

Regulating Data Transparency in Social Media

Chapter 03

As large social media companies have come under increasing global scrutiny, regulatory questions over transparency and access to public-interest data have gained prominence. Crucially, global regulatory and governance frameworks now recognise that platforms will not voluntarily provide the data necessary for independent, high-quality public-interest research (Coalition for Independent Technology Research, 2025).

In the EU, transparency requirements for UGC and advertising platform data are explicitly addressed in the Digital Services Act (DSA). Beyond publicly accessible data, the DSA establishes an unprecedented framework to grant vetted researchers access to certain categories of non-public platform data, including internal documentation and aggregated or otherwise privacy-preserving information relevant to the assessment of systemic risks (Scott & Seiling, 2025). While this mechanism constitutes an important regulatory precedent, its practical outcomes remain to be evaluated over time.

Advertising transparency regulation has, too, undergone a significant shift under the DSA. Previously, regulatory frameworks focused on requiring visible labels for users targeted by online ads, clarifying the commercial and promotional nature of the communication. Regulation now also addresses data access policies designed to enable independent scrutiny of advertisers' targeting, content, and persuasive strategies, going beyond platform self-regulatory measures (Santini et al., 2025a; United Nations, 2024).

In contrast, neither Brazil nor the United Kingdom currently have a dedicated legal framework for data transparency but have partially addressed this topic through adjacent regulations. In Brazil, the only provisions enabling researchers to access platform data stem from specific decisions and regulations issued by the country's Electoral Court (Santini et al., 2026). In the UK, the Online Safety Act (OSA) lacks specific provisions for independent research and scrutiny and is more focused on a risk-assessment framework.

The jurisdictions analysed in this study allow for a comparison of online platform data transparency across diverse regulatory contexts. The EU exemplifies a highly developed regulatory framework, offering a mature debate that often serves as a benchmark for other regions. The UK presents a distinctive approach—sometimes leveraging EU-inspired measures, yet at other times pursuing independent regulatory paths. Brazil, representing the Global Majority, illustrates a context where such debates are still emerging, regulatory frameworks are underdeveloped, and transparency practices largely rely on voluntary commitments. By comparing these cases, we capture varying levels of regulatory maturity and derive insights to inform policy development in regions where platform regulation remains nascent.

European Union

The EU serves as a prominent example of far-reaching and comprehensive regulation for data transparency in social media. The Digital Services Act (DSA) establishes a robust legal framework defining obligations for digital services, aiming to make digital spaces safer and fairer, protect users' fundamental rights, and ensure a level playing field for businesses. It came into force in November 2022 and became widely applicable across the EU from February 2024, with even earlier compliance deadlines for the largest platforms (European Commission, 2024a).

Notably, the DSA singles out Very Large Online Platforms (VLOPs) and Very Large Online Search Engines (VLOSEs), defined as those with more than an average of 45 million monthly active users in the EU, and imposes enhanced transparency and accountability obligations upon them (European Commission, 2025a). Among these obligations is the provision of data for vetted researchers, enabling independent scrutiny and monitoring of so-called systemic risks. Under the DSA, these risks include the dissemination and amplification of illegal, harmful, and misleading content, as well as its algorithmic curation and recommendation, alongside broader societal harms affecting fundamental rights, civic discourse, and public security (European Centre for Algorithmic Transparency, 2025).

Data access obligations are set out in Article 40 of the DSA, entitled "Data Access and Scrutiny". Article 40 is possibly the most far-reaching example of legally-mandated platform data transparency regulation worldwide. It sets out two separate mechanisms for research-related data access in paragraphs 12 and 4.

Paragraph 12 of Article 40 of the DSA states that eligible researchers who meet a set of criteria, including those affiliated with not-for-profit organisations, may obtain access to publicly accessible data on the online interfaces of VLOPs and VLOSEs. These conditions include: not having any commercial interests; disclosing the sources of research funding; demonstrating the ability to comply with data security and confidentiality requirements; and only accessing data that are strictly necessary for carrying out research for detecting, identifying, and understanding systemic risks (European Centre for Algorithmic Transparency, 2025). While well-intentioned, in practice, this framing means that, instead of public APIs, researcher access to public UGC data is often mediated through special programs for researchers and platform-designed application processes that create unnecessary hurdles and burdens, including restrictive eligibility criteria (Marsh & Seiling, 2026; Scott & Seiling, 2025). The increasing transfer of power to the hands of platforms remains a substantial risk, as platforms retain discretionary control over data access while benefiting from state endorsement (Mimizuka et al., 2025).

Paragraph 4 of Article 40 of the DSA states that researchers who meet a separate set of conditions may apply

for access to non-public data held by VLOPs and VLOSEs. These conditions include being affiliated with a recognised research institution, conducting research aimed at the detection, identification, and understanding of systemic risks and/or the assessment of risk mitigation measures disclosed by the platforms, and committing to making their research results publicly available free of charge (European Centre for Algorithmic Transparency, 2025).

This access is, in principle, operationalised through the recently established DSA Access Portal, through which eligible researchers may submit their data access requests. Access to data may occur either through direct transmission to vetted researchers via a dedicated interface or data storage system, or through a secure processing environment operated by the data provider or a trusted third party. In the latter case, researchers are granted access to a virtual environment rather than raw data (European Centre for Algorithmic Transparency, 2025).

Article 40 of the DSA was fully operationalised in July 2025 through a delegated act adopted by the European Commission. The act aims to address inconsistencies in its implementation, including differing application procedures and lengthy exchanges between researchers and platforms. This new legal device further clarifies how data access obligations for VLOPs and VLOSEs should work in practice, particularly regarding access for vetted researchers conducting public-interest research (European Commission, 2025b).

Although the delegated act implementing Article 40 does not explicitly address data scraping, it has been argued that, insofar as publicly accessible data from VLOPs must be made available for research purposes, scraping may be interpreted as a legitimate method when official access mechanisms—such as APIs—are insufficient or ineffective (Bekavac & Mayer, 2026). In this sense, the delegated act could be read as indirectly legitimising the use of scraping, reducing researchers' exposure to retaliation or legal uncertainty (Keller, 2025a; Marsh & Seiling, 2026). It has further been argued that such guarantees should prevail over platforms' terms of service, which have historically restricted or prohibited scraping activities (Keller, 2025b). However, the lack of standardised access mechanisms continues to generate uncertainty and, in some cases, hinders researchers' ability to obtain the data necessary for their work (Giglietto & Terenzi, 2025).

Recently, companies such as TikTok, Meta, and X/Twitter have been found in breach of their obligation to grant researchers adequate access to public data (European Centre for Algorithmic Transparency, 2025), which highlights persistent shortcomings in transparency requirement enforcement (Mimizuka et al., 2025). Moreover, as noted by the Commission's inquiry into X/Twitter, imposing access fees may unlawfully restrict researchers' ability to use publicly accessible data under the DSA (Marsh & Seiling, 2026).

Article 39 of the DSA, in turn, establishes specific data transparency requirements for advertising on online platforms. Under the regulation, platforms must ensure that all commercial communications are clearly identifiable to users and provide them with information on the advertiser and sponsor identities, the period during which the ad was promoted, and the targeting criteria used. In addition,

VLOPs and VLOSEs are required to maintain comprehensive ad repositories, making them accessible to users, researchers, and regulators through web-based tools and APIs for scrutiny and oversight. According to the DSA, these repositories must not rely on artificial distinctions between types of content, such as political versus non-political ads, but rather should allow scrutiny of all ads served on these platforms for up to one year after their last display on the platforms' online interfaces.

The repositories must protect the anonymity of users exposed to the ads while providing the ad content—whether video, image, or text—along with identifying information about the advertiser, the entity that paid for the ad (if different), the period of promotion, the targeting parameters used, and the total number of users reached. Nevertheless, these efforts continue to face significant limitations, particularly with respect to data completeness and comparability (Mozilla Foundation, 2024).

Complementing this framework, the EU's Regulation on Transparency and Targeting of Political Advertising (TTPA) introduces additional obligations specifically for political ads, including stricter requirements on targeting and the information provided in advertising repositories. In response to these rules, major platforms such as Meta and Google have announced plans to stop serving paid political and social issue ads in the EU, citing operational challenges and legal uncertainty (Kroeber-Riel, 2024; Meta, 2025). This led to certain transparency policies being discontinued, specifically for ads labelled as political on these platforms—most notably, Google went further by removing access to its political advertising record history and data in Europe (Votta, 2025).

Brazil

Brazil currently lacks a regulatory framework for establishing robust transparency obligations or meaningful mechanisms for accessing UGC data for independent researchers. Internet and digital governance are primarily shaped by the Brazilian Civil Rights Framework for the Internet (Marco Civil da Internet, Law No. 12.965/2014), which established foundational principles for internet use, privacy, net neutrality, and intermediary liability. Nevertheless, this legal framework does not fully guarantee the transparency of data from social media platforms, whether for public UGC or advertising.

More recently, data transparency and platform accountability mechanisms were proposed through the so-called "Fake News Bill", officially titled as the Brazilian Law on Freedom, Responsibility and Transparency on the Internet (Bill No. 2630/2020), which sought to regulate social media companies and combat the spread of toxic and malicious content online (Nazareno & Pinheiro, 2023). The bill, which was approved by the Federal

Senate in 2020, was heavily influenced by the DSA and also substantially revised after the DSA came into force in the EU (Bueno & Canaan, 2024).

However, it failed to advance to a plenary vote in the Chamber of Deputies and was ultimately shelved in April 2024, effectively halting its legislative progress. This outcome was largely driven by lobbying efforts and attempts to shape public opinion by online platform companies (Silva & Norigami, 2026). Google and Meta, for instance, consistently opposed the bill. Google even used its search engine to display a warning to users, claiming that the proposed law could change the internet as Brazilians knew it for the worse. The company also invested in a paid advertising campaign across its own platforms, as well as on Meta's platforms and even on Spotify, which allegedly does not allow political advertising. Telegram similarly sent messages to its Brazilian users with an alarmist tone, warning them about the supposed risks presented by the bill (Salles et al., 2025).

This gap also undermines the effectiveness of other recent institutional developments in the country. For instance, the country's Supreme Court has recently ruled that the Brazilian Civil Rights Framework for the Internet should no longer be interpreted as limiting platform liability to cases of non-compliance. Instead, the Court signaled the need for a more streamlined notice-and-takedown approach in certain circumstances, with a view to reducing reliance on lengthy judicial procedures. While this shift increases the potential for platforms to be made liable for unlawful content circulating on their services, it does not simultaneously establish corresponding transparency, oversight, or auditability mechanisms, leaving platforms subject to heightened responsibility without structurally enhanced public scrutiny, which in turn makes it more difficult to monitor compliance and enforce the law and its corresponding penalties.

Significant advances have occurred in advertising data, although a comprehensive regulatory framework is still lacking. In 2024, Brazil's Electoral Court, the constitutional authority for regulating electoral procedures, introduced new rules requiring those platforms that do allow the distribution of political and electoral advertising to maintain publicly accessible ad repositories that must also be accompanied by APIs (Tribunal Superior Eleitoral, 2024). These repositories must provide basic functions such as keyword search, and key information, including the amount spent on each ad and the size of the audience reached.

However, these measures produced unintended consequences. Leading platforms operating in Brazil, including Google and Kwai (Kuaishou), announced that they would cease to allow political and electoral advertising in the country rather than comply with the new transparency obligations (Nóbrega, 2024; Waltenberg, 2024). Despite this alleged prohibition in place, political and electoral ads have not been completely prevented from circulating on these platforms, and in fact often appear without meaningful constraints and without the transparency mechanisms envisioned by the Electoral Court (Santini et al., 2024).

Furthermore, Brazil recently passed the Digital Statute for Children and Adolescents (ECA Digital, Law No. 15.211/25), which came into force in March 2026. It is the country's first law

to regulate online platforms in any form, introducing measures for age verification, parental control, and the protection of minors from harmful online content. The law also establishes new transparency requirements, such as the periodic publication of transparency reports by major platforms, but does not include any provisions for data access (Câmara dos Deputados, n.d.).

United Kingdom

Unlike the far-reaching transparency framework and formalised mechanisms for researcher access to platform data established under EU's DSA, the United Kingdom's regulatory approach remains more constrained and, in the words of UK law, "proportionate". The country's primary framework for online platform regulation, the Online Safety Act 2023 (OSA), focuses on online content regulation and on protecting children and adults from harmful content (UK Parliament, 2023). The Act received Royal Assent in October 2023 and has been implemented gradually since 2024 (Department for Science, Innovation and Technology, 2025).

The OSA centres on establishing a statutory duty of care for online platforms, enforced by the UK's communications regulator, Ofcom. The framework empowers Ofcom to require platform risk assessments, issue codes of practice, and compel the disclosure of information necessary for investigating online harms. While the OSA includes certain transparency and accountability provisions, these are primarily directed toward regulators and designated experts rather than the general public, including public-interest researchers (UK Parliament, 2023).

The Data (Use and Access) Act 2025 subsequently introduces a significant, though still indirect, development regarding researcher access to platform data in the UK. The Act, which received Royal Assent in June 2025, amends the OSA by empowering the Secretary of State to establish a framework through secondary legislation that requires providers of regulated online services to disclose information to independent researchers for the purpose of conducting research on online safety issues (UK Parliament, 2025). While this provision appears similar to Article 40 of the DSA, it does not immediately create a comparable, fully functioning independent researcher access regime; rather, it delegates authority to future regulations, leaving the UK considerably behind the EU in establishing structured mechanisms for researcher access to social media data.

This regulatory deficit extends to online advertising. While all ads in the UK are required to be "legal, decent, honest and truthful" (Conway, 2025), regulation remains split between broadcast and non-broadcast media. Political advertising illustrates this divide. Under the Communications Act 2003 (UK Parliament, 2003), this type of advert is prohibited on broadcast media. However,

regulating authorities largely exempt political ads in non-broadcast media, including online platforms, from the content standards that apply to other types of advertising.

Originally enacted in 1999, this framework is insufficient to address the challenges posed by the dynamics of information and disinformation on social media platforms (Conway, 2025). Weak attempts to increase transparency around ad sources, such as the Elections Act 2022 (UK Parliament, 2022), have failed to regulate content or enable meaningful public scrutiny. Without legislation requiring social media platforms to provide free and universal access to digital advertising data, the UK remains particularly vulnerable to political online manipulation and advertising fraud.

Materials & Methods

Chapter 04

This study adopts a comprehensive framework grounded in data quality principles to evaluate researchers' access to public-interest data and to generate empirical evidence on whether, and under what conditions, platforms make such data available.

The assessment focuses specifically on publicly accessible data, defined as data derived from content which is accessible to any user of an online platform without requiring privileged or restricted access.

Scope of Analysis

The analysis focuses on publicly accessible social media UGC and advertising data for public-interest, not-for-profit research, auditing, and accountability purposes. Social media advertising is treated as inherently public-facing, as it is designed for broad dissemination through paid promotion. Public UGC, while also publicly accessible, is approached with greater ethical sensitivity due to its contextual and potentially personal nature. Accordingly, all data were handled in line with established ethical guidelines, including considerations of privacy, contextual integrity, and minimising potential harm (Association of Internet Researchers, 2019; Gliniecka, 2023; Lauterwasser & Nedzhvetskaya, 2023).

The study examines official mechanisms through which platforms systematically provide access to granular, post-level data, rather than aggregated or summary outputs. These mechanisms include, primarily, official APIs, as well as graphical user interfaces (GUIs) with data access functionalities. The analysis excludes unofficial scraping methods (Bruns, 2019; Freelon, 2018), third-party data brokers (Yadav & Wanless, 2022) and platform-curated datasets provided upon request (Morten et al., 2024; Windwehr & Selinger, 2024). Similarly, other forms of platform transparency, such as content moderation transparency reports, are not considered (Access Now, 2024; Urman & Makhortykh, 2023).

A total of 15 social media platforms, listed in **Table 1**, are assessed. In each case, platforms are evaluated against a set of weighted criteria, from which a composite score is derived. The analysis spans three different regulatory environments: the EU—where access to public-interest data for research is governed by the DSA— as well as Brazil and the United Kingdom, which serve as analytically relevant comparative cases. Where no regional differences in data access mechanisms were identified, a single assessment was conducted and the resulting score applied across all relevant jurisdictions.

Table 1

Plataforms Analysed and Data Access Mechanisms Considered for User-Generated and Advertising Content

Platform	Owner Company	UGC data access	Advertising data access	DSA VLOP Status[7]
Bluesky	Bluesky PBLLC	Bluesky Developer APIs	N/A	No
Discord	Discord Inc.	N/A	N/A	No
Facebook	Meta Platforms Inc.	Meta Content Library	Meta Ad Library	Yes
Instagram	Meta Platforms Inc.	Meta Content Library	Meta Ad Library	Yes
Kwai (Kuaishou/ SnackVideo)	Kuaishou Technology	N/A	N/A	No
LinkedIn	Microsoft Corporation	Beta Researcher Access Program	LinkedIn Ad Library	Yes
Pinterest	Pinterest Inc.	N/A	Pinterest Ads Repository	Yes
Reddit	Reddit Inc.	Reddit Research API	N/A	No
Snapchat	Snap Inc.	N/A	Snapchat Ads Gallery	Yes
Telegram	Telegram Messenger Inc.	Telegram APIs	N/A	No
Threads	Meta Platformas Inc.	Meta Content Library	Meta Ad Library	No
TikTok	ByteDance Ltd.	TikTok Research API	TikTok Commercial Content Library	Yes
X (Twitter)	X Corp.	X API	X Ads Repository	Yes
WhatsApp	Meta Platforms Inc.	N/A	Meta Ad Library	Yes
YouTube	Alphabet Inc. (Google)	YouTube Data API	Google Ads Transparency Center	Yes

[7] List of designated VLOPs and VLOSES available at https://digital-strategy.ec.europa.eu/en/policies/list-designated-vlops-and-vloses.

Scoring Criteria

Our framework is based on a set of assessment criteria structured around data quality dimensions, which enables an evaluation of not only whether data is made available, but also of how it is provided. Higher-weighted special criteria account for 75% of the total score, while the remaining criteria account for 25%. For UGC data, four special criteria and 25 additional criteria are considered; for advertising data, the framework comprises three special criteria and 33 additional criteria.

Table 2 presents each of the special criteria used in the assessment. The special criteria examine aspects such as API availability, scope, cost, and whether there is a GUI for data access. APIs stand out as a key transparency mechanism, enabling greater customisation of the data collection processes by allowing it to be automated and scaled, while also facilitating independent testing to verify whether extracted data correspond to what was intended. Crucially, APIs must support the independent extraction of structured data to users' own infrastructure, thereby upholding core open science principles and enabling researchers to verify results and collaborate across institutions (Bekavac & Mayer, 2026; Davidson et al., 2023; Wilkinson et al., 2016). GUIs also play an important role in democratising access for researchers and citizens with more limited technical resources or expertise (Giglietto & Terenzi, 2025).

Table 2

Special Assessment Criteria for Public User-Generated Content and Advertising Data Transparency Mechanisms

User-Generated Content Data Transparency	Advertising Content Data Transparency
Does the platform provide an API that enables the structured extraction of public user-generated content data for independent analysis?	Does the platform provide an API to access its ad repository and extract data on advertising content for independent analysis?
Can the full scope of public content data be extracted through the platform's API?	Does the platform provide a graphical user interface to its ad repository for extracting advertising content data?
Is access to the platform's API free of charge?	Can data from both active and inactive ads be extracted?
Does the platform offer a graphical interface for extracting data?	

The dimensions of data quality that underpin the other criteria include accessibility, compliance, completeness, consistency, relevance, timeliness, and accuracy. *Accessibility* ensures that the intended data can be easily discovered and retrieved by users. *Compliance* refers to adherence to widely adopted standards, applicable regulatory frameworks, and the documentation of the platform's data access mechanisms. *Completeness* verifies whether the data include all the information necessary to conduct the intended analyses. *Consistency* refers to the uniformity of data across all instances. *Relevance* requires that the data are meaningful and appropriately scoped for their intended uses. *Timeliness* indicates that the data are up to date and available when needed. Finally, *Accuracy* refers to the extent to which the data faithfully reflect the real-world entity or event they represent. Further details on the criteria under each dimension are provided online[8].

The final score, on a scale from 0 to 100, reflects the extent to which a platform enables independent, systematic, and high-quality data access for research purposes. Each platform–region assessment is then classified into one of the following transparency levels:

Meaningful (81-100)	The platform provides well-established, openly accessible data infrastructure—including free APIs, comprehensive documentation, and broad data coverage—enabling systematic content collection and monitoring with minimal barriers.
Limited (61-80)	The platform offers functional data access tools, but notable limitations—such as paywalls, restricted API scope, incomplete documentation, or access limited to approved researchers —require workarounds for comprehensive monitoring.
Deficient (41-60)	The platform provides partial transparency resources, but significant gaps in data coverage, filtering capabilities, and documentation prevent reliable, large-scale research.
Minimal (21-40)	The platform offers only minimal data access, with most transparency features either absent or severely constrained, hindering systematic monitoring.
Negligible (1-20)	The platform offers negligible transparency infrastructure, with nearly all criteria unmet and any existing tools too limited to support meaningful research.
Not Available (0)	The platform provides no data access mechanisms, despite the framework being applicable.

[8] Available at https://netlab.eco.ufrj.br/transparency

Results

Chapter 05

Our results show a fragmented landscape of data transparency across regions and platforms. Large platforms such as Meta, TikTok and LinkedIn concentrate meaningful transparency in jurisdictions with stronger oversight. The disparity is even more pronounced in advertising data, where the regional access gap is particularly stark. Differences also appear across platforms: while Bluesky and YouTube provide relatively strong transparency resources, demonstrating that meaningful access is technically feasible, many platforms offer little or no access even where regulatory requirements exist.

Social media transparency remains poor across both UGC and advertising. Our findings show that widely used platforms such as Discord and Kwai (Kuaishou) offer no data transparency mechanisms for either content type in any of the regions assessed. Meanwhile, major platforms, including those owned by Meta, and LinkedIn, offer only Negligible or Minimal levels of UGC data transparency across all regions, at best. Other platforms, such as Pinterest and Snapchat, provide only Negligible advertising transparency in a limited number of regions. Across the 15 platforms assessed in all regions, ten were rated as at best Deficient in terms of transparency resources for public UGC data, while at least seven platforms received similarly low ratings for advertising data transparency. This means that their transparency resources are either non-existent or subject to severe restrictions that hinder their effective use, such as limited API scope, significant gaps in data coverage, and missing data or filtering capabilities.

Platforms that lead in public UGC transparency offer little or no advertising transparency, and vice versa. Bluesky, which does not have an advertising-based business model, and YouTube are the only platforms classified as Meaningful for UGC data transparency, but YouTube provides only Deficient advertising transparency on average. X, Reddit, and Telegram also present a similar pattern of asymmetry between the two dimensions, favouring UGC data. Conversely, despite their low UGC data transparency scores, Meta's platforms (Facebook, Instagram, Threads, and WhatsApp) lead in advertising transparency in regulated markets.

Regulation appears to drive social media data transparency, especially in advertising data, but does not guarantee it. Under the DSA, the EU exhibits greater transparency yet still struggles to ensure compliance. X and Snapchat both fall under DSA requirements in the EU, yet provided only Negligible advertising transparency tools at the time of assessment. Although such tools formally exist, they are non-functional and fail to produce results, amounting to merely performative transparency.

The companies' compliance-driven approach toward transparency results in regional asymmetries. Brazil consistently records the lowest overall transparency scores for both public UGC and advertising data, while the EU achieves the highest scores for advertising transparency. The UK often matches or even exceeds EU levels, which may reflect regulatory spillover effects, whereby functionalities such as Meta Ad Library's features initially available only in the EU, as well as the availability of TikTok's Research API and Commercial Content Library, have been extended to the UK. While Meta's platforms, Google (YouTube), TikTok, and LinkedIn provide Limited or Meaningful transparency tools in the EU, they rate as only

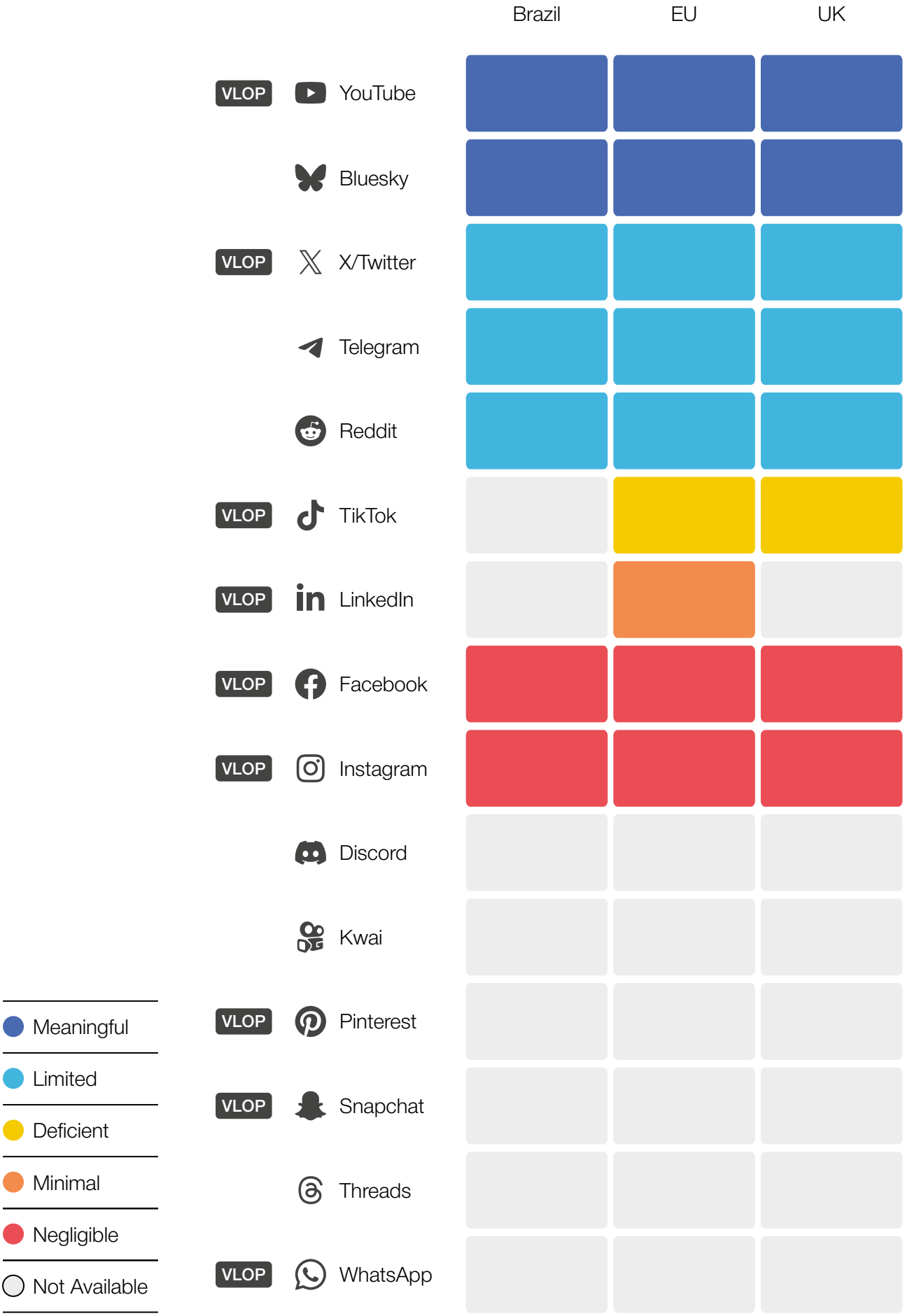
Deficient, Negligible, or Not Available in Brazil. This selective transparency, largely concentrated in jurisdictions with stronger regulatory oversight, produces significant disparities in the production of data-driven evidence on societal issues online.

Research capacity on social media is currently severely constrained across both public UGC and advertising data, making it nearly impossible to construct a comprehensive view of the contemporary information ecosystem. The ability to trace how content flows across platforms, how advertising targets audiences, and how information operations exploit the interplay between organic and paid content remains out of reach. Current fragmented data access is not a technical limitation but a pattern of selective disclosure shaped by corporate strategy and jurisdictional compliance. The stark divide between a small number of top performers (Bluesky and YouTube for UGC data, and Meta's platforms for advertising data in regulated markets) and the vast majority of services shows that data transparency is ultimately a matter of corporate choice, rather than a technical, commercial, or legal constraint.

Transparency cannot be achieved through mandates alone without accompanying standardisation of data access protocols and enforcement measures. Enforcement is only as strong as the data available to verify compliance, and where transparency tools are broken or absent, regulators lack the basic means to audit platform practices. Even under the DSA, platforms such as X, Snapchat, and Pinterest provide only Negligible transparency tools, with X having been fined by the European Commission for non-compliance in 2025 (Marsh & Seiling, 2026). The absence of effective penalties or enforcement mechanisms in many jurisdictions further compounds the problem, creating an environment in which non-compliance carries little cost.

The lack of common data access standards means that even well-intentioned regulatory frameworks produce fragmented outcomes. Each platform implements its obligations differently, forcing researchers to navigate a patchwork of incompatible systems, each with its own limitations and restrictions. This lack of interoperability not only raises the cost and complexity of research but also creates systematic advantages for well-resourced institutions in jurisdictions where platforms offer better data access, deepening existing inequalities in global research capacity. For researchers in the Global Majority, who often face the most acute information integrity challenges, these barriers are especially consequential.

User-Generated Content Data Transparency Assessments



Despite regulatory progress, access to public UGC data remains limited in all regions. Ten of the 15 platforms assessed either do not have any interface or programmatic means for data access, or only provide access at a level we classified as, at best, Deficient across all assessed regions. Discord, Kwai, Pinterest, Snapchat, Threads, and WhatsApp lack any data transparency resource and have therefore been deemed as Not Available. Similarly, transparency on Facebook and Instagram has been classified as Negligible across all regions, as extracting structured data from the Meta Content Library (MCL) API to a users' devices violates the platforms' terms and conditions. TikTok only provides a level of Deficient access in the EU and the UK, while LinkedIn offers only Minimal access in the EU.

Apart from these few exceptions, the EU's greater regulated environment does not appear to provide a substantial advantage in public UGC data transparency compared to the UK and Brazil. In practice, our request for access to LinkedIn's Research API under Article 40 of the DSA was not approved, and we were only able to rely on its public documentation to assess a limited set of data quality criteria. Similarly, our request for access to X's API under Article 40 of the DSA was denied. Pinterest and Snapchat also do not provide any programmatic access under Article 40 of the DSA. Instead, they offer curated datasets to selected researchers under their own conditions, which do not comply with the standards outlined in the *Materials & Methods* section.

Meta has drastically reduced access to public UGC data across its platforms. CrowdTangle, its previous solution for UGC data transparency, offered relatively flexible access but was discontinued in 2024. It was replaced by the MCL, a far more restricted and controlled environment comprising an API and a GUI, to which our access was approved and which we were able to test for this study. However, the MCL API was classified as Negligible, as its terms and conditions prohibit the extraction of structured data to users' own devices. Through the MCL API, pre-approved users can access and analyse data from Instagram and Facebook only, and solely within a controlled-access environment. By contrast, the MCL GUI allows the extraction of structured raw data from Facebook and Instagram, albeit with significant limitations on the availability of public data. Threads data can be accessed through this interface but cannot be extracted in any form. For WhatsApp, no public UGC data (e.g. from public channels) is accessible via either the MCL API or its GUI.

Ranked in an intermediate position, Reddit and X offer reasonably developed data transparency resources, although both present critical limitations that undermine research efforts. Both platforms were classified as having Limited transparency for UGC, meaning they provide structured access to data but with constraints on volume and/or data quality. As with Meta's shift from CrowdTangle to the MCL, the transition from Twitter to X marked a significant regression in data transparency in the early 2020s. Once widely regarded as a leading example of openness in social media data access—offering a free, well-documented API extensively used for research—X introduced paid access to its API in 2023 (Mehta, 2024). At the time of our evaluation in 2025, monthly subscription costs started at USD 200 and

could reach up to USD 5,000 for more extensive research use[9].

Our request to access X's API under the provisions of the DSA in the EU was initially submitted in October 2025 and, after multiple follow-up attempts, was definitively rejected in December of the same year, without further justification. As a result, the API could not be effectively used or assessed under these conditions. This experience is consistent with that reported by other researchers and contributes to a broader pattern that ultimately led to the imposition of a fine against X in the EU due to its inadequate transparency practices (Marsh & Seiling, 2026).

Among the messaging platforms evaluated, Telegram is the only one that offers any data transparency mechanisms for public UGC. Through its API, Telegram provides free programmatic access to messages in public channels and groups, scoring highly across our criteria for accessibility, completeness, consistency, and timeliness. The main limitations of data access on Telegram are the absence of platform-wide content discovery—requiring prior knowledge of specific channels—and shortcomings in documentation quality, which place it in the **Deficient** band of our analysis. Discord and WhatsApp offer no mechanisms for researchers to access or extract public UGC data.

In the UK and the EU, TikTok performs moderately in our analysis thanks to its Research API, which is available to researchers in the U.S. and Europe, but not in Brazil. TikTok's free API access, robust documentation, and data filtering capabilities place it in the **Deficient** band under our methodology. However, its application and approval process, combined with delays in updating the data made available to researchers, create a misalignment between what is observable on the platform and what is accessible for critical inquiry and research, ultimately limiting its overall performance. Although TikTok is not subject to regulatory obligations to provide this API to UK-based researchers, it has done so, which we hypothesise to be a form of regulatory spillover—part of the "Brussels Effect", whereby regulatory standards adopted in the EU shape platform practices beyond its jurisdiction (European Parliament, 2025; Husovec & Urban, 2024).

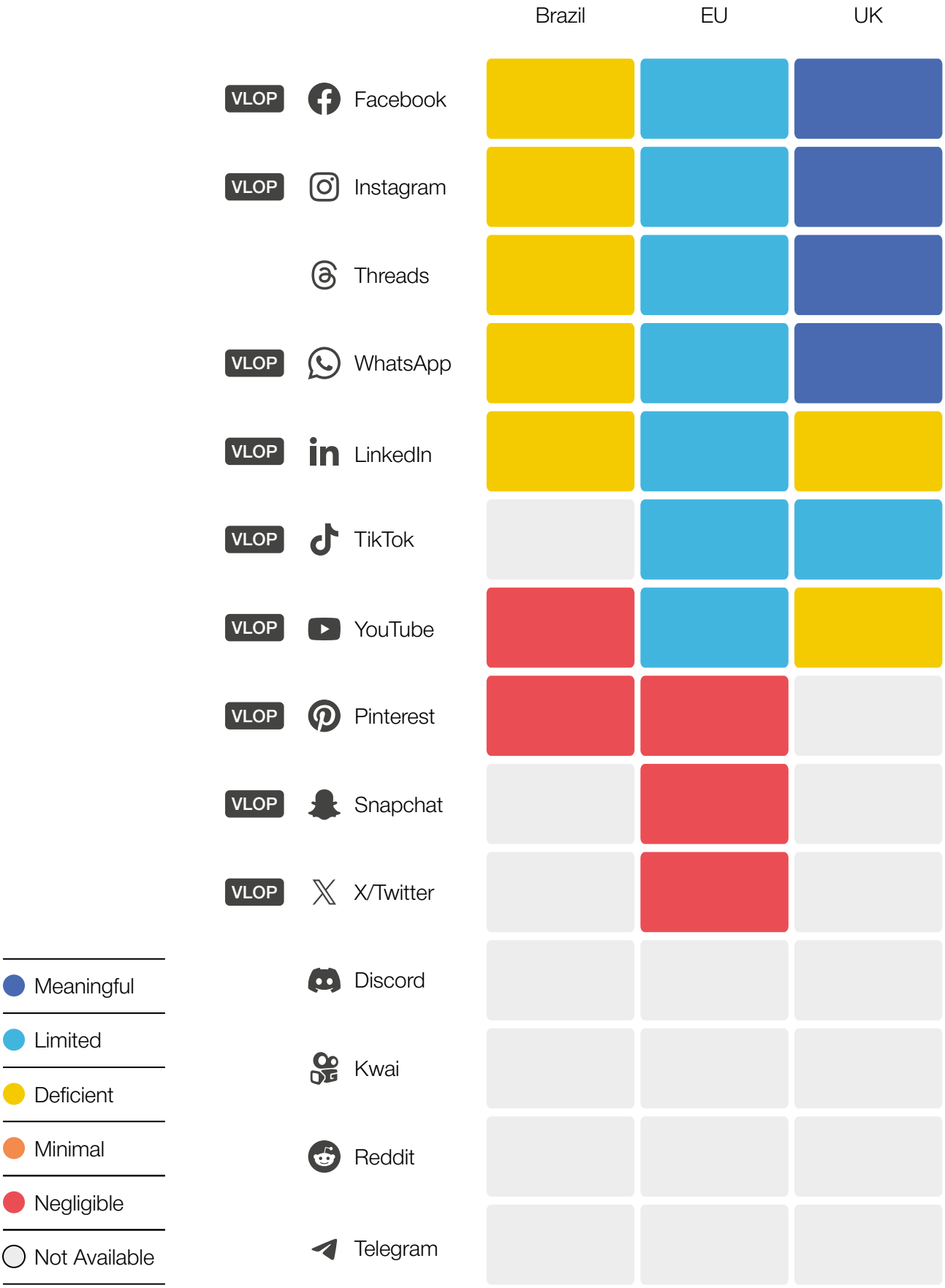
Bluesky and YouTube emerge as the top performers in the transparency assessment for public UGC data, with both offering free, full-scope programmatic access to such data. Bluesky's open-source architecture is designed to enable transparency and unrestricted data access, although its documentation received a weaker evaluation. YouTube, in turn, offers a well-structured API with extensive documentation; however, strict quota limits significantly limit access in practice, undermining transparency for large-scale research. Overall, these are the only two platforms classified in the **Meaningful** band in our analysis.

---

[9] Following our evaluations conducted between late 2025 and early 2026, X announced and began rolling out a new API access model based on a pay-per-use subscription whereby users, researchers, and developers are charged according to the volume of data they extract (Singh, 2026). Although the new pricing model has lower entry barriers, it still hinders independent research in low-resource or cost-sensitive environments.

Even on platforms that provide some degree of transparency, critical bottlenecks continue to constrain social media research. Essential functionalities such as native geographic and language-based filtering are often absent from APIs, while access to historical content remains negligible or not available. As a result, researchers conducting localised analyses must extract substantially larger datasets than necessary and rely on computationally intensive post-processing. These approaches can introduce additional sources of bias and error, while also requiring significant infrastructure—constraints that systematically disadvantage under-resourced researchers, often those best positioned to investigate information integrity challenges in underserved regions and languages. At the same time, restricted access to historical data precludes longitudinal analysis, limiting the ability to examine how phenomena such as disinformation campaigns or hate speech evolve over time. These limitations are not marginal inconveniences, but fundamental constraints on the kinds of knowledge that can be produced about the digital information environment.

Advertising Content Data Transparency Assessments



Half of the platforms assessed for advertising transparency were classified as Not Available or Negligible across all regions. Overall, seven of the 14 platforms with an advertising business model fail to meet basic transparency standards—Bluesky was not evaluated as it does not run ads. Discord, Kwai, Reddit, Telegram, and Pinterest provide no mechanisms for structured access to advertising data. X and Snapchat maintain ad transparency tools in the EU under DSA obligations, but these were non-functional at the time of assessment, placing both in the Negligible band.

Advertising data transparency reflects a landscape shaped by both regulatory pressure and voluntary decisions. Unlike public UGC data—where platforms such as Bluesky, YouTube, X, Telegram, and Reddit provide Meaningful or Limited transparency for global access—advertising transparency is marked by pronounced regional disparities. Among platforms that offer some level of ad transparency, the gap between Brazil and the UK and the EU is both stark and consistent. Meta's platforms achieve a Meaningful classification in the UK but drop to Deficient in Brazil; YouTube declines from Limited in the EU to Negligible in Brazil; and TikTok falls from Limited in the EU to Not Available in Brazil. These patterns indicate that the technical infrastructure exists but is deployed selectively.

Despite the positive effects of regulation, some companies subject to the DSA lack meaningful transparency and were classified as providing Negligible resources in the EU, highlighting the gap between nominal compliance and functional transparency. Pinterest, X, and Snapchat exemplify this pattern. In the EU, our application to X's API access to advertising data remained pending without response after three months and could therefore not be evaluated, while its GUI export feature returned no data across all tested queries, which are limited to advertiser selection. In Brazil and the UK, X was classified as Not Available, with no transparency tools. Similarly, Snapchat's EU ad transparency tools proved non-functional: the API consistently returned errors, and the GUI yielded no results regardless of the search queries applied. Pinterest, in turn, offers a GUI-based ad library in the EU and Brazil, but not in the UK, allowing users to browse ads and view basic advertiser and temporal information. However, Pinterest provides no API or structured data extraction capabilities and does not support keyword search, comprehensive filtering, or systematic analysis. A previously available API for advertising data extraction was deprecated in 2022 and has not been replaced.

The highest levels of advertising transparency were observed on Meta's platforms in the United Kingdom and the European Union, with the UK scoring slightly higher. Facebook, Instagram, Threads, and WhatsApp all rely on the Meta Ad Library as their primary transparency resource and offer broadly similar features across both regions. The main difference is that Meta allows the extraction of political advertising data in the UK, whereas in the EU, political advertising has reportedly been banned to avoid compliance with the TTPA regulation (Votta, 2025). As a result, the data export feature is not available through the GUI for any ad type in the EU. Because this criterion is heavily weighted in our methodology, Meta scored comparatively higher in the UK. In the EU, its ad data transparency was classified as Limited, alongside platforms

such as YouTube (Google), TikTok, and LinkedIn. Nonetheless, the overall scores for advertising transparency in the EU and the UK remain relatively close.

The artificial distinction between political and non-political ads also reduces access to Meta advertising data in Brazil. There, Meta's ad transparency is mostly restricted to content labeled as political or related to social issues. By contrast, in the EU and the UK, the Ad Library API provides comprehensive access to data on all ads delivered to users, including both active and inactive ads, with information on advertisers, funding sources, audience demographics (such as age and gender), impressions, temporal data, and moderation actions. In Brazil, however, both the API and the GUI limit structured data extraction to political and social issue ads, while other ads can only be viewed through the interface but not exported. The absence of statutory transparency requirements for all advertisements in Brazil weakens the enforcement of consumer protection, child protection, and electoral regulations, among others.

Google's approach to advertising transparency illustrates a compliance-driven model that fragments access across jurisdictions. In the EU, the company provides comprehensive programmatic access to data on all active and inactive ads served over the past year, including advertiser information, verification status, moderation flags, and delivery dates, broadly mirroring the Ads Transparency Center interface. As the GUI's data export is restricted to ads labelled as political—and such ads were banned in the EU following the entry into force of the TTPA—this functionality is no longer available in the region. In the UK, data export features, whether through the GUI or the API, are limited to political advertising, while commercial ads can only be viewed through the interface with more limited information, placing the platform in the Deficient band. In Brazil, Google's mid-2024 ban on political advertising, motivated by new norms introduced by the country's Electoral Court, has effectively eliminated structured access to up-to-date ad data, whether via the GUI or the API, while restricting visibility to commercial ads from verified advertisers only, a limitation not present in the EU or UK, placing the platform in the Negligible band.

Across all regions, Google's structured advertising data lacks ad content, audience demographic breakdowns, and granular spending information. More critically, both programmatic data access and the Ads Transparency Center GUI are constrained by the inability to locate ads through keyword searches, as queries can only be performed based on the identification and selection of advertisers. This design effectively requires researchers to already know who advertised in order to retrieve information, significantly hindering data discovery and raising concerns that the system operates as a form of transparency washing (Zalnieriute, 2021).

The TikTok Commercial Content Library is another example of regulatory spillover from which the UK benefits, as, although it was introduced to comply with the DSA's requirements, it also includes data on ads served to users in the country. While its GUI is publicly accessible to users worldwide, it only includes ads served to users in Europe. API access is restricted to approved researchers, providing structured data on advertiser identity, funding sources, targeting criteria, and delivery dates. However, rate limits are undocumented,

spending and engagement data are unavailable, and the interface does not support structured data export. Overall, this places TikTok's advertising data transparency in the Limited band in the UK and the EU, while remaining unavailable in other regions.

LinkedIn stands out as the only platform to offer global API access to its ad library across all ad types, although significant regional limitations remain. While the API performs more robustly for ads served in the EU—where LinkedIn was classified as Limited—it provides incomplete data for ads delivered outside the region. Notably, the API does not return ad creative content in any jurisdiction, offering only the ad URL alongside basic advertiser and metadata. In addition, both the ad library and the API fail to disclose key information such as ad delivery dates and targeting criteria, although these data points are available for EU-served ads.

Several advertising transparency features available in the UK point to the presence of regulatory spillover effects. Functionalities of the Meta Ad Library originally restricted to ads served in the EU have been extended to the UK, and the TikTok Commercial Content Library is likewise accessible in the country. None of these tools is required under any UK regulatory framework, but the UK is the only market outside the EU to be significantly affected by such DSA-related decisions and measures. However, even this spillover is asymmetrical and only extends to a few platforms, underscoring the need of equivalent domestic guarantees to secure and broaden transparency.

The design of ad repository search functionalities has implications that shape the possible forms of accountability. When platforms require users to search by advertiser name—rather than by ad content, topic, or targeting criteria—the architecture itself becomes a mechanism of opacity. Identifying fraudulent, misleading, or harmful ads becomes effectively impossible without prior knowledge of the advertiser's identity. The result is an architecture that favors browsing and exploratory use over systematic, data-driven investigation. This limitation is particularly consequential in the context of influence operations, paid disinformation campaigns, and microtargeted political advertising, where the harm lies in the granular dynamics of audience selection and message delivery that are not captured by aggregated data.

The opacity of advertising data has direct consequences for the oversight of electoral campaigns and democratic governance. Without the ability to independently verify the volume, content, targeting, and spending of political ads on social media platforms, electoral authorities, researchers, and civil society cannot meaningfully assess whether campaigns are operating within legal boundaries. This undermines the capacity to ensure fairness and equal opportunity among candidates. In jurisdictions such as Brazil, platforms have responded to regulatory pressure by banning political advertising without extending transparency obligations to advertising more broadly—unlike in the EU, where such bans coexist with comprehensive, regulated transparency requirements. This asymmetry compounds the problem: political ads continue to circulate without oversight, effectively rendering electoral advertising regulation unenforceable.

The lack of standardised advertising data across platforms is a fundamental barrier to effective sector governance.

Variations in format, granularity, coverage, and definitions make consistent cross-platform comparison nearly impossible, limiting regulators' ability to assess the ecosystem holistically and constraining the production of reliable comparative research. Without interoperable standards, platforms operate in isolation, under distinct disclosure practices, undermining efforts to ensure information integrity across the digital environment.

Advertising opacity also undermines both advertisers' commercial interests and platform credibility. Digital advertising is central to the business models of major social media companies and is projected to reach a global market value of USD 650 billion in 2025 (Precedence Research, 2025). Yet transparency and independent oversight remain limited. In the absence of independent verification of ad delivery and performance, investment decisions rely on unauditable, platform-generated metrics—"magical numbers" that are widely cited but fundamentally unverifiable. This erodes budget accountability, distorts competition, and weakens confidence in the online advertising marketplace. When performance claims cannot be independently verified, advertisers are more likely to direct budgets toward ineffective or inflated advertising products.

Recommendations

Chapter 06

The findings of the *Social Media Data Transparency Index* confirm that meaningful transparency in social media data remains the exception rather than the norm. Addressing this critical issue requires coordinated international action from multiple stakeholders.

The following recommendations are organised by stakeholder groups and guided by two key imperatives: ensuring transparency of public UGC and advertising data to support public-interest research, and addressing institutional gaps that perpetuate regional and platform-level asymmetries.

Social Media Companies

R1

Ensure meaningful universal access to all advertising data

Social media platforms should provide free, programmatic, and open access to their full advertising datasets through both APIs and GUIs. Advertising repositories should contain the complete database of ad data, disclosed at the individual ad level, and should allow for exports with multiple filters. Access should require no more than a standard developer registration process, and platforms should avoid application-based procedures that selectively restrict transparency.

These repositories must meet robust quality standards, ensuring that data is complete, consistent, accurately filterable, and available with sufficient historical depth. All ads served on a platform should be subject to the same transparency standards, on the grounds that advertising is inherently public-facing and therefore a matter of public interest. Furthermore, platforms should not remove historical advertising data from their repositories, as archive data are essential for longitudinal research, electoral oversight, and the identification of patterns in influence operations that only emerge over time.

Critically, this means that access should not differentiate based on ad content or purpose. Our findings show that, even where ad repositories exist, they often impose significant constraints, such as artificial distinctions between political and non-political advertising. We argue that it is not possible to draw clear or widely accepted boundaries between these categories. Reliance on advertiser self-classification is also insufficient, as content may be mislabelled, whether intentionally or not.

To fulfil their purpose, ad repositories should enable searches and data extraction across multiple parameters, including keywords, topics, targeting criteria, spending ranges, time periods, and geographic reach. Their design must support systematic, data-driven investigation rather than merely exploratory browsing.

Tools that restrict searches to advertiser names function, in practice, as mechanisms of opacity, as identifying fraudulent, misleading, or harmful ads becomes nearly impossible without prior knowledge of the advertiser. This limitation is particularly acute in the context of influence operations, paid disinformation campaigns, and microtargeted political advertising, where risks stem from the granular dynamics of audience targeting and message delivery that aggregated data fail to capture.

R2

Ensure meaningful universal access to public user-generated content data for public-interest uses

Platforms should provide programmatic, free access to public user-generated content (UGC) data that meets clear standards of completeness and quality, made available through both APIs and GUIs to accommodate for different levels of technical capacity and expertise. Access should be open and non-discretionary, and take place outside controlled or secure environments, enabling independent data extraction and analysis. At the same time, the use of this data should be governed by clear, non-binding terms and conditions that ensure adherence to ethical standards, including respect for privacy, contextual integrity, and the mitigation of potential harm.

Access should not depend on platform approval, unreasonable fees, or restrictive environments that hinder independent research. Distinctions between "eligible" and "non-eligible" actors risk generating large numbers of unjustifiably denied requests. We therefore propose the adoption of standardised terms of access and use agreements defining the technical and ethical obligations of all parties involved. Upon acceptance of these terms, access should remain in full compliance with applicable data protection frameworks and restrictions on the commercial use of data.

R3

End selective and fragmented data transparency practices

Social media companies should align their global UGC and advertising data access practices with the most transparent regulatory standards, rather than relying on minimal or jurisdiction-specific compliance. Selective transparency, often shaped by uneven regulatory regimes, creates significant disparities in the global capacity for data-driven research and democratic oversight. Regions facing some of the most acute information integrity challenges—particularly across much of the Global Majority—are often those where platforms provide the least access. There is no legitimate technical or commercial justification for offering lower levels of transparency in some regions than in others.

Our findings show that platforms tend to deploy more transparent practices in jurisdictions with stronger regulatory frameworks, while maintaining more limited access elsewhere. Companies should instead align their data access practices to the highest available standard, ensuring that users and researchers everywhere benefit from consistent levels of transparency.

R4

Provide vetted researchers with access to non-public data within the limits defined by democratic regulation and oversight

Although this aspect falls outside the scope of our assessment framework, access to non-public data—as regulated under Article 40 of the DSA—is essential for monitoring platform-enabled systemic risks and online harms, as well as for independently auditing platforms' self-reported transparency metrics. Beyond publicly accessible data, platforms hold non-public datasets that are critical for understanding risks such as coordinated disinformation campaigns, algorithmic curation, and internal content moderation practices.

Platforms should therefore provide secure, controlled research environments, such as cleanrooms and research sandboxes, and work with democratic national and regional authorities to implement robust researcher-vetting processes and clear access protocols that ensure both data protection and meaningful research use. These environments should enable qualified independent researchers to analyse sensitive datasets under strict security conditions, with access calibrated to the level of data sensitivity.

Such infrastructures should also support independent verification of platform claims in transparency reports and public communications. The current practice of publishing performance metrics, content moderation statistics, and risk assessments without external validation undermines public trust and weakens the credibility of both social media platforms and the broader online information ecosystem.

R5

Provide access to data on moderated and removed content

Rather than erasing traces of abusive, harmful, or even illegal practices, social media platforms should maintain auditable databases covering both UGC and advertising content that has been moderated or removed. This data should be made available to enable the reconstruction of threats to information integrity and to assess how platforms are addressing and enforcing moderation actions, including the scope and limits of such interventions. It would also allow for comparisons between removed and non-removed content, enabling more in-depth and systematic assessments of moderation practices.

Moderated and removed advertising data should follow the principle of universal access, given that advertising content is inherently public-facing, and should therefore be included in the same ad repositories as non-moderated content. Sensitive UGC data, however, should at a minimum be accessible within secure research environments, subject to appropriate safeguards for privacy and data protection. This approach ensures that the evidence records of platform interventions remain available for independent scrutiny, rather than being permanently erased alongside the content itself.

International Governance Bodies

R6

Advance the information integrity agenda through data access and transparency standards

The implementation of data access should move from self-regulated, voluntary commitments to legally binding regulatory frameworks that enable effective democratic oversight.

As the UN and other international organisations have recognised, transparency and research access to data are essential to safeguarding information integrity. This recognition

must now be translated into operational and enforceable mandates.

While voluntary codes of conduct remain important, our *Transparency Index* highlights that self-regulation has structural limitations when ensuring consistent and meaningful access. In this context, multistakeholder bodies, independent organisations, and their relevant agencies should take a more active and coordinated role in establishing and promoting international principles and standards governing the technical, ethical, and operational dimensions of data access, thereby informing transnational policymaking. Civil society, academic networks, and international organisations should also work collectively to advance data transparency and information integrity as shared global priorities.

Our findings point to a fragmented landscape: even in regulated regions such as the EU, platforms implement transparency mechanisms in inconsistent and often incompatible ways, forcing researchers to navigate patchwork systems that increase costs, introduce inconsistencies, and disadvantage under-resourced institutions, particularly in the Global Majority. International bodies responsible for online communication standards should therefore lead efforts to develop and endorse interoperable protocols for publishing UGC and advertising data, drawing on established data quality frameworks and incorporating input from researchers to ensure that transparency mechanisms produce data that is not only available, but also reliable, comparable, and fit for analytical use.

Regional Regulation

R7

European Union: Fully enforce Articles 39 and 40 of the Digital Services Act

The DSA stands out as the most advanced regulatory framework for platform data transparency worldwide, yet significant challenges in implementation and enforcement remain. Most social media platforms are still not fully compliant with its regulations for advertising and UGC data access. Even under Article 39, many platforms fail to provide functional advertising repositories with APIs, often offering only GUIs that return no results. Under Article 40, independent researchers, including ourselves, frequently have data access requests denied with no justification.

The European Commission should ensure effective implementation through standardised access protocols and be prepared to impose meaningful penalties, including substantial fines and, where necessary, temporary suspension of services, in the case of non-compliance. Article 40 should be fully operationalised to guarantee that vetted researchers receive timely and continuous access to relevant platform data, including advertising repositories, content dissemination metrics, and systemic risk indicators.

The remit of Article 40(12) must also be clarified to prevent restrictive interpretations that undermine its purpose. Researcher access should not be mediated through cumbersome platform-designed application processes, restrictive eligibility criteria, or other mechanisms that give platforms undue control over data provision. In addition, the scope of "publicly accessible data" under Article 40(12) should be interpreted broadly. Strong enforcement backed by effective and dissuasive penalties is essential to bridge the gap between the law on paper and transparency in practice.

As we hypothesise, doing so will have effects well beyond the EU's borders, driven by the dynamics of the "Brussels Effect". The DSA continues to serve as a benchmark for platform regulation worldwide, shaping both emerging and more established frameworks. This influence is broadly positive, but only insofar as the model it provides proves effective in practice.

R8

Brazil: Consolidate existing legal principles into a dedicated transparency framework for platform data access

Brazil should draw on lessons from the DSA's to develop a dedicated framework for social media data transparency, guaranteeing access to public UGC and advertising data. This framework should consolidate principles from existing laws—including the Data Protection Law (13.709/2018), the Consumer Defence Code, and the Child and Adolescent Statute (15.211/2025)—while also incorporating insights from prior legislative debates, such as Bill 2630/2020, through a multistakeholder governance approach.

Brazil currently lacks robust transparency obligations and consistently scores the lowest in our assessments, particularly for advertising data. The shelving of Bill 2630/2020 left the country with no comprehensive platform governance legislation. The consequences are tangible: platforms such as Google and Kwai have banned political advertising rather than comply with Electoral Court transparency requirements, yet evidence suggests that such ads continue to circulate without oversight, effectively

undermining the enforceability of electoral advertising rules.

Advertising transparency in Brazil should therefore be grounded in consumer protection and competition law. Consumers have a well-established right to know the nature, origin, and reach of the content they are exposed to. Incorporating data access provisions into the consumer protection framework—through instruments such as an executive decree or ministerial order—would be a practical and immediately actionable step, with significant potential to improve transparency in one of the world's largest online advertising markets.

R9

United Kingdom: Move from case-by-case oversight toward a dedicated transparency framework for platform data access

The UK should adopt a proactive transparency framework guaranteeing access to online platform data, drawing on lessons from the EU's DSA while strengthening enforcement and standardisation mechanisms. While the Online Safety Act 2023 establishes regulatory oversight of social media companies, the current approach relies largely on government-led supervision and case-specific interventions.

Our findings show that UK transparency scores often match or even exceed EU levels for Meta's platforms and TikTok, mostly because companies voluntarily extend DSA-compliant tools to the UK market. This dynamic, apparently a result of the "Brussels Effect" described in the literature, provides some benefits but offers no domestic guarantees, leaving UK researchers and institutions at a regulatory disadvantage compared to their EU counterparts.

The Data (Use and Access) Act 2025 introduces the possibility of a future researcher access framework but does not immediately establish a regime comparable to Article 40 of the DSA. Advertising transparency obligations, which remain fragmented and outside a comprehensive governance framework, should be a priority within any new domestic approach.

R10

Brazil and the United Kingdom: Clarify the legality and ethical use of web scraping for public-interest research

Following the DSA's provisions on publicly available data, Brazilian and UK authorities should clarify and explicitly support the legal and ethical use of web scraping for public-interest research. Although the delegated act implementing Article 40 does not explicitly address scraping, it has been argued that, insofar as publicly accessible data from VLOPs must be made available for research purposes, this provision can be interpreted as legitimising scraping where official access mechanisms are inadequate.

In practice, scraping often occurs under uncertain and adverse conditions, exposing researchers to legal risks and potential platform retaliation. Given that platform-provided APIs and interfaces frequently fail to deliver meaningful or complete access, the legal status of alternative data collection methods has become a matter of practical urgency. Public-interest access guarantees should take precedence over restrictive terms of service that have historically limited independent data collection. Regulators should therefore issue clear guidance to reduce legal uncertainty and the chilling effects that deter research, while ensuring that data collection practices remain aligned with data protection requirements and ethical standards.

Public-Interest Researchers and Institutions

R11

Promote international research consortia and foster interdisciplinary collaboration

Researchers and their institutions should pool knowledge, methodologies, and technical resources through multinational, cross-sector, and interdisciplinary collaboration. They should actively support efforts for regulation and standardisation by contributing technical expertise, documenting access failures, and working across regions to strengthen the

evidence base for platform governance. The *Social Media Data Transparency Index* and its accompanying report offer a modest illustration of this approach.

Researcher engagement is essential not only for producing knowledge but also for demonstrating the practical value of data access and sustaining political will for enforcement. The collective experience that researchers have gathered from facing barriers when accessing much-needed data to study the societal impacts of technology in and of itself constitutes an important evidence base that should inform regulatory design and enforcement priorities.

Researchers should actively participate in consultations on data access standards, contribute to the assessment of transparency mechanisms, and foster cross-regional partnerships to address the geographic inequalities identified in our findings. This also constitutes a call to funding bodies and organisations to prioritise and support transnational collaborative projects, as disparities in resources and capacities are among the primary barriers to their development. Such collaboration should extend beyond academia to include civil society, journalists, and other relevant stakeholders, ensuring that its outcomes translate into broader societal benefits—such as the formulation of new public policies, the strengthening of existing regulatory frameworks, and the promotion of more accountable market practices.

R12

Develop monitoring strategies for low-access settings

Researchers should develop free and open-source software and design technical and methodological approaches that are easily deployable in contexts where data access is severely restricted and resources are limited. Complementary monitoring strategies—including browser-based measurement tools, cloud-based data pooling, crowdsourced data donations, and other independent auditing methods—can help generate evidence on platform dynamics even when official channels are blocked or dysfunctional.

These resources can simultaneously strengthen local research capacity and broaden the scope of demands for data transparency and information integrity. At the same time, researchers should document methodological limitations and access constraints transparently, which would contribute not only to the immediate evidence base but also to the broader case for improved regulatory frameworks that would render such workarounds unnecessary.

Bibliography

Access Now. (2024). *Transparency Reporting Index*. https://www.accessnow.org/campaign/transparency-reporting-index/

Angus, D., Hayden, L., Obeid, A. K., Tan, X. Y., Carah, N., Burgess, J., Parker, C., Andrejevic, M., Fordyce, R., Cellard, L., & Bagnara, J. (2024). Computational methods for improving the observability of platform-based advertising. *Journal of Advertising*, *53*(5), 661-680. https://doi.org/10.1080/00913367.2024.2394156

Association of Internet Researchers. (2019). *Internet research: Ethical guidelines 3.0*. https://aoir.org/reports/ethics3.pdf

Batini, C., & Scannapieco, M. (2016). *Data and information quality*. Springer. https://link.springer.com/book/10.1007/978-3-319-24106-7

Bekavac, L., & Mayer, S. (2026). *Auditing Meta and TikTok research API data access under Article 40(12) of the Digital Services Act*. arXiv preprint. https://doi.org/10.48550/arXiv.2601.12390

Benkler, Y., Farris, R., & Roberts, H. (2018). *Network propaganda: Manipulation, disinformation, and radicalization in American politics*. Oxford University Press. https://doi.org/10.1093/oso/9780190923624.001.0001

Benson, P. R. (2019). *The guide to ISO 8000*. ECCMA. https://eccma.org/forms/Publications/download_library_data/VGhlIEd1aWRlIHRvIElTTyA4MDAw/V2hpdGVwYXBlcg==

Blake, T., Nosko, C., & Tadelis, S. (2015). Consumer Heterogeneity and Paid Search Effectiveness: A Large-Scale Field Experiment. *Econometrica*, *83*(1), 155–174. https://doi.org/10.3982/ECTA12423

Bradshaw, S., & Howard, P. N. (2019). *The global disinformation order: 2019 global inventory of organised social media manipulation*. Oxford Internet Institute. https://demtech.oii.ox.ac.uk/wp-content/uploads/sites/12/2019/09/CyberTroop-Report19.pdf

Bruns, A. (2019). After the 'APIcalypse': Social media platforms and their fight against critical scholarly research. *Information, Communication & Society*, *22*(11), 1544–1566. https://doi.org/10.1080/1369118X.2019.1637447

Bueno, T. M., & Canaan, R. G. (2024). The Brussels Effect in Brazil: Analysing the impact of the EU Digital Services Act on the discussion surrounding the fake news bill. *Telecommunications Policy*, *48*(5), 102757. https://doi.org/10.1016/j.telpol.2024.102757

Câmara dos Deputados. (n.d.). *ECA Digital: mais proteção para crianças e adolescentes*. https://infograficos.camara.leg.br/eca-digital/

Cavale, S. (2018, March 1). P&G says cut digital ad spend by $200 million in 2017. *Reuters*. https://www.reuters.com/article/business/pg-says-cut-digital-ad-spend-by-200-million-in-2017-idUSKCN1GD653/

Chandlee, B. (2019, October 3). Understanding our policies around paid ads. *TikTok Newsroom*. https://newsroom.tiktok.com/understanding-our-policies-around-paid-ads

Check My Ads Institute. (2025, January). *Digital Advertising and Its Role in the 2024 Southport Riots*. https://checkmyads.org/wp-content/uploads/2025/03/Digital-Advertising-

and-Its-Role-in-the-2024-Southport-Riots_20250225-web.pdf

Coalition for Independent Technology Research. (2023). *Letter: Imposing fees to access the Twitter API threatens public-interest research*. https://independenttechresearch.org/letter-twitter-api-access-threatens-public-interest-research/

Coalition for Independent Technology Research. (2025). *Power in Numbers*. https://independenttechresearch.org/wp-content/uploads/2025/08/The-State-of-Independent-Technology-Research-Power-in-Numbers.pdf

Conway, L. (2025, June 25). *Regulation of advertising in the UK*. House of Commons Library. https://commonslibrary.parliament.uk/research-briefings/sn06130/

Davey, J., & Ebner, J. (2019). *'The great replacement': The violent consequences of mainstreamed extremism*. Institute for Strategic Dialogue. https://www.isdglobal.org/publication/the-great-replacement-the-violent-consequences-of-mainstreamed-extremism/

Davidson, B. I., Wischerath, D., Racek, D., Parry, D. A., Godwin, E., Hinds, J., van der Linden, D., Roscoe, J. F., Ayravainen, L., & Cork, A. G. (2023). Platform-controlled social media APIs threaten open science. *Nature Human Behaviour*, 7(12), 2054–2057. https://doi.org/10.1038/s41562-023-01750-2

Department for Science, Innovation and Technology. (2025). *Online Safety Act: Explainer*. GOV.UK. https://www.gov.uk/government/publications/online-safety-act-explainer/online-safety-act-explainer

Edelson, L. (2020). *Facebook's political ad spending numbers don't add up*. Medium. https://medium.com/online-political-transparency-project/transparency-theater-facebooks-political-ad-spending-numbers-don-t-add-up-d7a85479a002

European Centre for Algorithmic Transparency. (2025). *FAQs: DSA data access for researchers*. European Commission. https://algorithmic-transparency.ec.europa.eu/news/faqs-dsa-data-access-researchers-2025-07-03_en

European Commission. (2018). *Tackling online disinformation: A European approach*. https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:52018DC0236

European Commission. (2024a, February 22). *Questions and answers on the Digital Services Act*. https://ec.europa.eu/commission/presscorner/detail/en/qanda_20_2348

European Commission. (2024b, December 4). *Commission, online platforms and civil society increase monitoring during Romanian elections*. https://ec.europa.eu/commission/presscorner/detail/en/ip_24_6243

European Commission. (2025a). *DSA: Very large online platforms and search engines*. https://digital-strategy.ec.europa.eu/en/policies/dsa-vlops

European Commission. (2025b). *Delegated act on data access under the Digital Services Act (DSA)*. https://digital-strategy.ec.europa.eu/en/library/delegated-act-data-access-under-digital-services-act-dsa

European Parliament. (2018, October 25). *The use of Facebook users' data by Cambridge Analytica and the impact*

on data protection. https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:52018IP0433

European Parliament. (2024). *Information integrity online and the European democracy shield*. https://www.europarl.europa.eu/RegData/etudes/BRIE/2024/767153/EPRS_BRI%282024%29767153_EN.pdf

European Parliament. (2025). *Spillover effect of the Digital Services Act (DSA) into non-EU countries*. https://www.europarl.europa.eu/doceo/document/E-10-2025-003271_EN.html

European Partnership for Democracy. (2020). *Universal advertising transparency by default*. https://algorithmwatch.org/en/wp-content/uploads/2020/09/joint-call-for-universal-ads-transparency.pdf

Fou, A. (2021, January 2). *When Big Brands Stopped Spending On Digital Ads, Nothing Happened. Why?*. Forbes. https://www.forbes.com/sites/augustinefou/2021/01/02/when-big-brands-stopped-spending-on-digital-ads-nothing-happened-why/

Freelon, D. (2018). Computational research in the post-API age. *Political Communication*, *35*(4), 665-668. https://doi.org/10.1080/10584609.2018.1477506

Giglietto, F., & Terenzi, M. (2025). *The state of social media research APIs & tools in the Digital Services Act era*. Università degli Studi di Urbino Carlo Bo Repository. https://ora.uniurb.it/handle/11576/2759111?mode=simple

Gliniecka, M. (2023). The ethics of publicly available data research: A situated ethics framework for Reddit. *Social Media + Society*, *9*(3). https://journals.sagepub.com/doi/10.1177/20563051231192021

Henley, J. (2024, November 25). *Shock as pro-Russia independent wins first round of Romanian election*. The Guardian. https://www.theguardian.com/world/2024/nov/25/calin-georgescu-romania-election-hard-right-candidate

Horwitz, J. (2025, November 6). *Meta is earning a fortune on a deluge of fraudulent ads, documents show*. Reuters. https://www.reuters.com/investigations/meta-is-earning-fortune-deluge-fraudulent-ads-documents-show-2025-11-06/

Husovec, M., & Urban, J. (2024, February 21). *Will the DSA have the Brussels Effect?*. Verfassungsblog. https://verfassungsblog.de/will-the-dsa-have-the-brussels-effect/

Joseph, J. (2025, June 6). *Elon Musk's X Blocks Others from Using Its Data for AI Training*. PCMag. https://www.pcmag.com/news/elon-musks-x-blocks-others-from-using-its-data-for-ai-training

Keller, D. (2025a, October 27). *Using the DSA to study platforms*. Verfassungsblog. https://verfassungsblog.de/dsa-platforms-digital-services-act/

Keller, D. (2025b, December 12). *How the meaning of "publicly accessible" shapes researcher data rights under the DSA*. Tech Policy Press. https://www.techpolicy.press/how-the-meaning-of-publicly-accessible-shapes-researcher-data-rights-under-the-dsa/

Kroeber-Riel, A. (2024, November 14). *An update on political advertising in the European Union*. Google Blog. https://blog.google/company-news/inside-

google/around-the-globe/google-europe/political-advertising-in-eu/

Lauterwasser, S., & Nedzhvetskaya, N. (2023, August 11). *Privacy in public? The ethics of academic research with publicly available social media data*. Berkeley Journal of Sociology. https://berkeleyjournal.org/2023/08/11/privacy-in-public/

Ledford, H. (2020, June 17). *How Facebook, Twitter and other data troves are revolutionizing social science*. Nature. https://www.nature.com/articles/d41586-020-01747-1

Leerssen, P., Ausloos, J., Zarouali, B., Helberger, N., & de Vreese, C. (2019). Platform ad archives: Promises and pitfalls. *Internet Policy Review*, *8*(4). https://doi.org/10.14763/2019.4.1421

Le Pochat, V., Edelson, L., Goethem, T. V., Joosen, W., McCoy, D., & Lauinger, T. (2022). An audit of Facebook's political ad policy enforcement. In *Proceedings of the 31st USENIX Security Symposium* (pp. 607–624). USENIX Association. https://www.usenix.org/conference/usenixsecurity22/presentation/lepochat

Mahanti, R. (2019). *Data quality: Dimensions, measurement, strategy, management, and governance*. ASQ Quality Press.

Marres, N. (2017). *Digital sociology: The reinvention of social research*. Polity Press.

Marsh, O., & Seiling, L. (2026, February 12). *What the X fine reveals about data access under Article 40 of the Digital Services Act*. Tech Policy Press. https://www.techpolicy.press/what-the-x-fine-reveals-about-data-access-under-article-40-of-the-digital-services-act/

Marwick, A., & Lewis, R. (2017). *Media manipulation and disinformation online*. Data & Society Research Institute. https://datasociety.net/library/media-manipulation-and-disinfo-online/

Mehta, I. (2024, October 30). *X makes its basic API tier more costly, launches annual subscriptions*. Tech Crunch. https://techcrunch.com/2024/10/30/x-makes-its-basic-api-tier-more-costly-launches-annual-subscriptions/

Meta. (2025, July 25). *Ending political, electoral and social issue advertising in the EU in response to incoming European regulation*. Meta Newsroom. https://about.fb.com/news/2025/07/ending-political-electoral-and-social-issue-advertising-in-the-eu/

Mimizuka, K., Brown, M. A., Yang, K.-C., & Lukito, J. (2025). *Post-post-API age: Studying digital platforms in scant data access times*. arXiv preprint. https://arxiv.org/abs/2505.09877

Morten, C. J., Nicholas, G., & Viljoen, S. (2024). Researcher access to social media data: Lessons from clinical trial data sharing. Berkeley Technological Law Journal, *39*, 109–204. https://btlj.org/wp-content/uploads/2024/04/0003_39-1_Morten.pdf

Mozilla Foundation. (2019). *Ad transparency report: European Union*. https://adtransparency.mozilla.org/eu/2019-05-17/eu/

Mozilla Foundation. (2024, April 16). *Stress testing tech platforms' ad repositories*. https://www.mozillafoundation.org/en/research/library/full-disclosure-stress-testing-tech-platforms-ad-repositories/

Murphy, H., Criddle, C., & Morris, S. (2025, October 22). *Reddit sues AI search engine Perplexity for scraping its data*.

Financial Times. https://www.ft.com/content/8a3da2aa-bfc1-4847-862d-978ff1ba24c9?syn-25a6b1a6=1

Nazareno, C., & Pinheiro, G. P. (2023, April). *Regulação de plataformas, fake news e o PL 2630/2020*. Câmara dos Deputados. https://bd.camara.leg.br/bd/bitstreams/0c7332c3-a181-4adb-8037-80dcd50e3e5b/download

Nóbrega, L. (2024, May 16). *Kwai veta anúncios políticos para as eleições de 2024*. desinformante. https://desinformante.com.br/kwai-anuncios-politicos/

Nuţu, O., Ioniţă, S., Pârvu, S., Voinea, M. (2025, January). EFOR Annual Report 2025: The crisis of the Romanian state. *Expert Forum*. https://expertforum.ro/wp-content/uploads/2025/01/Annual-report-2025-EFOR-ENG.pdf

OECD. (2019). *Enhancing access to and sharing of data*. https://www.oecd.org/en/publications/enhancing-access-to-and-sharing-of-data_276aaca8-en/full-report.html

Olari, V. (2024, December 12). *Rise of unknown Romanian presidential candidate preceded by Telegram and TikTok engagement spikes*. DFRLab. https://dfrlab.org/2024/12/12/romania-candidate-telegram-tiktok/

Pearson, G. D. H., Silver, N. A., Robinson, J. Y., Azadi, M., Schillo, B. A., & Kreslake, J. M. (2024). Beyond the margin of error: A systematic and replicable audit of the TikTok research API. *Information, Communication & Society*, *28*(3), 452–470. https://doi.org/10.1080/1369118X.2024.2420032

Pothong, K., Kaynak, S., Fry, D., Ghai, S., Livingstone, S., Phippen, A., Soriano, C. R., Given, L. M., Howard, P. N., Valente, M., & Valenzuela, S. (Eds.). (2026). *Disrupting digital exploitation: Recommendations for preventing and responding to technology-facilitated child sexual exploitation and abuse* (Summary for Policymakers, SFP2026.1). International Panel on the Information Environment. https://doi.org/10.61452/UYSB8827

Precedence Research. (2025). *Digital ad spending market size, share and trends 2026 to 2035*. https://www.precedenceresearch.com/digital-ad-spending-market

Rainsford, S. (2024, December 4). *Romania hit by major election influence campaign and Russian cyber-attacks*. BBC. https://www.bbc.com/news/articles/cgq18w507dko

Rieke, A., & Bogen, M. (2018, May). *Leveling the platform: Real transparency for paid messages on Facebook*. Upturn. https://www.upturn.org/static/reports/2018/facebook-ads/files/Upturn-Facebook-Ads-2018-05-08.pdf

Rogers, R. (2009). *The end of the virtual: Digital methods*. Amsterdam University Press.

Schiffrin, A., Marwick, A. E., Sinha, N., Wangnoo, A., Williams, K. Huseynova, E., & Hatfield, A. (2026). *Deepfake Financial Fraud: The Global Regulation of AI-Driven Scams*. Data & Society Research Institute and Columbia University Technology and Policy Innovation. https://datasociety.net/wp-content/uploads/2026/03/PolicyBrief_Deepfake-Financial-Fraud_March2026_typographic-final.pdf

Seers, P., Wilkes, T. R., and Horwitz, J. (2026, March 18). *Exclusive: Meta vowed to stop illegal financial ads in Britain. It failed 1,000 times in a week*. Reuters. https://www.reuters.com/sustainability/boards-policy-regulation/meta-vowed-stop-

illegal-financial-ads-britain-it-failed-1000-times-week-2026-03-18/

Salles, D., Mattos, B., Santini, R. M., & Canavarro, M. (2025). Unfair Play: Digital Platform's Abuse of Power to Influence Brazilian Policy Agenda. *AoIR Selected Papers of Internet Research*. https://doi.org/10.5210/spir.v2024i0.14053

Santini, R. M., Salles, D., Mattos, B., Canavarro, M., Haddad, J. G., Silva, D., & Loureiro, F. (2024, July 12). *Nota técnica: Google diminui transparência de anúncios políticos no Brasil e desobedece resolução do TSE*. NetLab UFRJ. https://netlab.eco.ufrj.br/post/nota-tecnica-google

Santini, R. M., Salles, D., Mattos, B., Moreira, A., Mello, D., Haddad, J. G., Fernandez, M. E., Dias, B., & Benzecry, L. (2025a). *Transparency under threat: Progress and setbacks between CrowdTangle and the Meta Content Library*. NetLab UFRJ. https://netlab.eco.ufrj.br/en/post/transparency-under-threat-progress-and-setbacks-between-crowdtangle-and-the-meta-content-library

Santini, R. M., Salles, D., Mattos, B., Sanchotene, N., & Belin, L. (2025b). *Atingido pelas redes sociais: Os impactos da indústria da desinformação nos consumidores brasileiros.* Editora Sulina. https://netlab.eco.ufrj.br/post/atingidos-pelas-redes-sociais-os-impactos-da-ind%C3%BAstria-da-desinforma%C3%A7%C3%A3o-nos-consumidores-brasileiro

Santini, R. M., Salles, D., Mattos, B., & Dau, E. (2026, February 3). *Propaganda eleitoral em 2026: recomendações à resolução nº 23.610/2019 e às novas minutas do Tribunal Superior Eleitoral (TSE)*. NetLab UFRJ. https://netlab.eco.ufrj.br/post/propaganda-eleitoral-em-2026

Scott, M., & Seiling, L. (2025, November 12). *In critical condition – How to stabilize researcher data access?*. Tech Policy Press. https://www.techpolicy.press/in-critical-condition-how-to-stabilize-researcher-data-access/

Silva, M., & Norigami, M. (2026, February 12). *Brazil: Big Tech cries 'censorship' and spreads disinformation*. SOMO. https://www.somo.nl/brazil-big-tech-cries-censorship-and-spreads-disinformation/

Singh, R. (2026, February 7). *X Revamps Developer API Pricing, Shifts to Pay-Per-Use Model*. MediaNama. https://www.medianama.com/2026/02/223-x-developer-api-pricing-pay-per-use-model/

Sosnovik, V., & Goga, O. (2021). Understanding the complexity of detecting political ads. In *Proceedings of the Web Conference 2021* (pp. 2002–2013). https://doi.org/10.1145/3442381.3450049

Srivastava, A. K., & Mishra, R. (2023). Analyzing social media research: A data quality and research reproducibility perspective. *IIM Kozhikode Society & Management Review*, *12*(1), 39–49. https://doi.org/10.1177/22779752211011810

Tribunal Superior Eleitoral. (2024). *Resolução n.° 23.732, de 27 de fevereiro de 2024*. https://www.tse.jus.br/legislacao/compilada/res/2024/resolucao-no-23-732-de-27-de-fevereiro-de-2024

Tromble, R. (2021). Where have all the data gone? A critical reflection on academic digital research in the post-API age. *Social Media + Society*, 7(1), 2056305121988929. https://doi.org/10.1177/2056305121988929

Tucker, J. A., Guess, A., Barbera, P., Vaccari, C., Siegel, A., Sanovich, S., Stukal, D., & Nyhan, B. (2018). *Social media, political polarization, and political disinformation: A review of the scientific literature*. SSRN. https://doi.org/10.2139/ssrn.3144139

Tufekci, Z. (2014). Big questions for social media big data: Representativeness, validity and other methodological pitfalls. *Proceedings of the International AAAI Conference on Web and Social Media*, *8*(1), 505-514. https://doi.org/10.1609/icwsm.v8i1.14517

UK Parliament. (2003). *Communications Act 2003*. https://www.legislation.gov.uk/ukpga/2003/21/contents

UK Parliament. (2022). *Elections Act 2022*. https://www.legislation.gov.uk/ukpga/2022/37/contents

UK Parliament. (2023). *Online Safety Act 2023*. https://www.legislation.gov.uk/ukpga/2023/50

UK Parliament. (2025). *Data (Use and Access) Act 2025: Explanatory notes*. https://www.legislation.gov.uk/ukpga/2025/18/notes/division/12/index.htm

United Nations. (2024). *United Nations global principles for information integrity – Recommendations for multi-stakeholder action*. https://www.un.org/sites/un2.un.org/files/un-global-principles-for-information-integrity-en.pdf

United Nations. (2025). *From principles to practice: Strengthening information integrity*. https://www.un.org/sites/un2.un.org/files/un-information-integrity-issue-brief-01.pdf

Urman, A., & Makhortykh, M. (2023). How transparent are transparency reports? Comparative analysis of transparency reporting across online platforms. *Telecommunications policy*, *47*(3), 102477. https://doi.org/10.1016/j.telpol.2022.102477

van Atteveldt, W., & Peng, T.-Q. (2018). When communication meets computation: Opportunities, challenges, and pitfalls in computational communication science. *Communication Methods and Measures*, *12*(2–3), 81–92. https://doi.org/10.1080/19312458.2018.1458084

Vosoughi, S., Roy, D., & Aral, S. (2018). The spread of true and false news online. *Science*, 359(6380), 1146-1151. https://doi.org/10.1126/science.aap9559

Votta, F. (2025, November 3). *What data reveals about Meta and Google's political ad ban in the EU*. Tech Policy Press. https://www.techpolicy.press/what-data-reveals-about-meta-and-googles-political-ad-ban-in-the-eu/

Waltenberg, F. (2024, April 23). *Google veta impulsionamento eleitoral em 2024*. Poder360. https://www.poder360.com.br/eleicoes/google-veta-impulsionamento-eleitoral-em-2024-e-pressiona-tse/

Wilkinson, M. D., Dumontier, M., Aalbersberg, I. J., Appleton, G., Axton, M., Baak, A., Blomberg, N., Boiten, J.-W., da Silva Santos, L. B., Bourne, P. E., Bouwman, J., Brookes, A. J., Clark, T., Crosas, M., Dillo, I., Dumon, O., Edmunds, S., Evelo, C. T., Finkers, R., Gonzalez-Beltran, A., Gray, A. J. G., Groth, P., Goble, C., Grethe, J. S., Heringa, J., 't Hoen, P. A. C., Hooft, R., Kuhn, T., Kok, R., Kok, J., Lusher, S. J., Martone, M. E., Mons, A., Packer, A. L., Persson, B., Rocca-Serra, P., Roos, M., van Schaik, R., Sansone, S.-A., Schultes, E., Sengstag, T., Slater, T., Strawn, G., Swertz, M. A., Thompson, M., van

der Lei, J., van Mulligen, E., Velterop, J., Waagmeester, A., Wittenburg, P., Wolstencroft, K., Zhao, J., & Mons, B. (2016). The FAIR guiding principles for scientific data management and stewardship. *Scientific Data*, *3*(1), 160018. https://doi.org/10.1038/sdata.2016.18

Windwehr, S., & Selinger, J. (2024, Mar 27). *Can we fix access to platform data? Europe's Digital Services Act and the long quest for platform accountability and transparency*. Internet Policy Review. https://policyreview.info/articles/news/can-we-fix-access-to-platform-data

Who Targets Me. (2025, May 22). *The disappearing ad library*. https://whotargets.me/en/the-disappearing-ad-library/

Woods, L., & Antoniu, A. (2025). *Advertising and the Online Safety Act 2023*. Online Safety Act Network. https://www.onlinesafetyact.net/documents/564/Advertising_and_the_Online_Safety_Act_2023.pdf

X. (n.d.-a). *X Developer Platform*. https://developer.x.com/

X. (n.d.-b). *Pricing - X API*. https://docs.x.com/x-api/getting-started/pricing

Yadav, K., & Wanless, A. (2022). *Social Listening Companies and Access to Sensitive Data*. Carnegie Endowment for International Peace. https://informationenvironment.org/wp-content/uploads/2022/10/PE2-Social-Listening-Companies-and-Access-to-Sensitive-Data.pdf

Zalnieriute, M. (2021). "Transparency Washing" in the Digital Age: A Corporate Agenda of Procedural Fetishism. Critical Analysis of Law, *8*, 139–153. https://doi.org/10.33137/cal.v8i1.36284

Zubiaga, A. (2018). A longitudinal assessment of the persistence of Twitter datasets. *Journal of the Association for Information Science and Technology*, *69*(8), 974–984. https://doi.org/10.1002/asi.24026

Terminology

Online platforms

Online platforms are technological infrastructures that connect individuals and organisations, facilitating the exchange of goods, services, or information by mediating relationships among two or more parties. They often include technical features enabling third parties to interact with the platform or develop additional functionalities. Such platforms are inherently non-neutral and operate under profit-driven incentives. While online platforms take various forms (e.g., app stores, streaming services, online marketplaces), this work focuses specifically on one type: *social media*.

Social media platforms

Social media platforms are a subset of online platforms that host and circulate user-generated content and interactions rather than producing most content themselves. In addition to user-generated content, these platforms feature advertising and commercial material, which form the basis of their business models. These models rely on computational infrastructures designed for large-scale collection, processing, and analysis of user data. Even though the content is largely created by users, platforms make crucial decisions about its distribution—determining what is shown, to whom, how user connections and interactions are structured, and what is permitted.

Messaging platforms

While primarily used for real-time, personal, and encrypted communication, messaging platforms (e.g., Discord, WhatsApp, Telegram) increasingly blur the boundaries between private and public interactions. They are progressively incorporating social components into their core features, such as large groups, public channels, status updates, and other broadcast-oriented or community-building functionalities, which facilitate the widespread circulation of content and the formation of networked publics. Consequently, we argue that messaging platforms should be considered an integral part of the social media ecosystem.

User-Generated content

Social media platforms are predominantly populated by two distinct types of content: user-generated content and advertising content. User-generated content (UGC) refers to any material created and shared by users on a platform—such as posts, comments, images, or videos—rather than by the platform itself, and distributed organically without paid promotion. For most platforms, UGC is not only a means of enabling users to communicate and connect, but also a core mechanism through which user engagement is transformed into behavioural data and, ultimately, economic value.

Advertising content

On the other hand, advertising content can be understood as any form of commercial communication whose visibility is amplified through payments made to the platform. This type of content is typically distributed in a microtargeted manner: drawing on the accumulation of users' behavioural, demographic, and geographic data—derived both from their interactions and from information they voluntarily provide—platforms construct detailed user profiles. These profiles are then made available to advertisers, who use them to target audiences they consider most likely to receive, engage with, and respond to the promoted content.

Application Programming Interface

Application Programming Interfaces (APIs) are among the most widely used methods for collecting structured data from social media platforms. They enable communication between software components—typically a server (such as a platform database) and a client (for instance, a researcher's device)—through standardised data requests governed by specific protocols. By allowing users to securely access and retrieve data in a structured and automated manner, APIs support scalable data collection processes and offer a high degree of customisation, as researchers can tailor queries to specific parameters and research needs. Access generally requires authentication tokens to validate client requests, while data retrieval occurs through calls to predefined endpoints, following parameters specified in the API's documentation and executed through code.

By facilitating data exchange, APIs promote interoperability across otherwise heterogeneous systems. However, some APIs adopt a more restrictive design, requiring prior approval and limiting access to secure and controlled environments. We argue that this model is not ideal for online platform-based research and inquiry, as it undermines core principles of open science by preventing independent verification of results, restricting cross-validation with other data sources, and constraining collaboration and data sharing across research communities.

Web scraping

Web scraping is a method for extracting and aggregating online content, often used by researchers when official data access mechanisms—such as APIs—are limited or insufficient. In practice, however, scraping is typically restricted by platform terms of service and actively discouraged through technical barriers, requiring continuous effort to maintain data collection and processing workflows. This dynamic not only increases the technical burden on researchers but also introduces legal uncertainty, as platforms have, in some cases, pursued action against individuals engaging in scraping.

While platforms justify these restrictions as necessary to protect user data, ensure authenticity, prevent automated abuse, and safeguard their ecosystems, scraping remains a far-from-ideal approach for social media research. It poses significant challenges related to data quality, reliability, and completeness, as well as to the long-term maintenance and stability of the infrastructures required to support data collection at scale.

Graphical User Interface

All platforms are accessed through a graphical user interface (GUI): a visual and interactive environment that allows users to perform actions and input commands in an intuitive, non-programmatic way. In addition to their primary interfaces, some platforms also provide dedicated interfaces for data access and exploration, enabling the collection and navigation of both UGC and advertising content.

User-Generated Content

In the case of UGC, a notable example is CrowdTangle, a tool developed by Meta that allowed approved researchers and journalists to browse and extract structured public data from platforms within its ecosystem. Data access could be performed either programmatically, via an API, or directly through the graphical interface, often with minimal technical effort. The tool also offered accessible and interactive features for data visualisation and analysis. CrowdTangle was eventually replaced by the Meta Content Library (MCL), introduced in 2023 as a private archive of public UGC from Facebook, Instagram, and Threads, with more restricted access and functionality. Access to the MCL is limited to researchers pre-approved by Meta, including those affiliated with academic institutions, research institutes, not-for-profit organisations, and other entities engaged in scientific or public-interest research. Data can be accessed either through a GUI with features similar to those previously offered by CrowdTangle or via an API. However, API access is only granted within a tightly controlled environment, typically through a closed virtual machine system that prevents data export and restricts integration with researchers' own infrastructures, thereby limiting data sharing and independent analysis.

Advertising Content

In the context of advertising, ad repositories are primarily structured as web-based layers accessible through a GUI, and may be accompanied by an API. These GUIs allow users to browse ads served on a platform over a given period, providing access to their content as well as metadata such as delivery dates, reach, targeting criteria, and spending. In some cases, platforms also enable the extraction of this information in structured formats through simple interface-based actions, although comprehensive data coverage is often not guaranteed. Ad repository APIs, in turn, allow for the structured extraction of the data made available through these GUIs. However, they frequently impose additional limitations, including restrictions on the categories of ads whose data can be accessed, thereby constraining the scope and completeness of independent analysis.

Data Not Found

net lab
Laboratório de Estudos de Internet e Redes Sociais

UFRJ
UNIVERSIDADE FEDERAL DO RIO DE JANEIRO

MINDEROO CENTRE FOR TECHNOLOGY & DEMOCRACY

UNIVERSITY OF CAMBRIDGE